

Theorems and Complete Proof

1 Joint Optimization Objective

In this section, we draw inspiration from redistribution theory in economics, and propose an objective function to optimize diversity and fairness while maintaining accuracy. We explain the diversity and fairness objectives separately, and then discuss some nice properties of our objective function.

Overall objective. As illustrated in the Introduction, diversity corresponds to the Sufficentarian principle while fairness corresponds to the Rawlsian principle. Drawing inspiration from this correspondence, we propose an objective function to optimize both diversity and fairness in recommender systems. We design the function to be piecewise, where diversity is optimized in the first stage and fairness becomes the target in the latter stage.

Let $\mathcal{I} = \{1, \dots, I\}$ denote the item set and let $I = |\mathcal{I}|$ denote the number of items. Vectors indexed by items are written in $\mathbb{R}^I$; sums over items are written either as $\sum_{i=1}^I$ or $\sum_{i \in \mathcal{I}}$.

Define $f: [0, 1]^I \rightarrow \mathbb{R}$ for utility vector $\mathbf{v} \in [0, 1]^I$ to be the full objective function:

$$f(\mathbf{v}) = \sum_{i=1}^I \mathcal{F}(v_i), \quad (1)$$

where $\mathcal{F}: [0, 1] \rightarrow \mathbb{R}$ is the function for the utility $v_i \in [0, 1]$ of a single item i :

$$\mathcal{F}(v_i) = \underbrace{\lambda C_\delta^*(v_i)}_{\text{diversity objective}} + \underbrace{U_\alpha^*(v_i)}_{\text{fairness objective}}, \quad (2)$$

where $C_\delta^*(\cdot)$ and $U_\alpha^*(\cdot)$ are diversity and fairness terms, respectively. $\mathcal{F}$ is a piecewise function divided into diversity stage and fairness stage, as detailed in Sections 1.1 and 1.2. $\delta \in (0, 1]$ is the coverage threshold to measure diversity. It is also the dividing point between two stages. $\alpha \in [0, \infty)$ is the iso-elastic fairness-aversion parameter: larger values give relatively more marginal weight to lower-utility items. Coefficient λ is to ensure the smoothness of the function at the dividing point, which we will discuss in Section 1.3.

1.1 Diversity Objective

In this section, we introduce how we design the diversity function. First, we define the diversity objective in the following form:

$$C_\delta^*(v) = \begin{cases} 1 - e^{-v/\delta}, & 0 \leq v < \delta, \\ 1 - e^{-1}, & \delta \leq v \leq 1. \end{cases}$$

which takes item utility v as input, and outputs how much it contributes to diversity with coverage threshold set to δ .

Our diversity function aligns with the Bass diffusion model with external influence [1], which models the growth of production adoptions. This function form incentivizes the surge of item utilities to reach the coverage threshold at the early stage of online recommendation, analogous to the Sufficentarian principle in redistribution economics. It "saturates" and falls into a constant when the item is already covered, which aligns with the definition of aggregate diversity. Concavity exhibits a diminishing return growth pattern and encourages breadth (exposing many items) over depth (heavily exposing a few items).

Diversity axioms in RS. To motivate the functional form, we first state axioms for the unsaturated pre-threshold coverage curve. Theorem 1 characterizes the unique continuous memoryless pre-threshold curve; the saturated objective used by DivFair is then imposed as a separate aggregate-coverage design choice.

Based on the concept of aggregate diversity and a rich line of related work [7, 8, 10], the unsaturated pre-threshold coverage family $\{C_\delta^{\text{pre}}\}_{\delta>0}$ is motivated by the following axioms.

A1 (Monotonicity). $(C_\delta^{\text{pre}})'(v) \geq 0, \forall v \geq 0$. More utility never *reduces* diversity.

A2 (Law of diminishing returns). $(C_\delta^{\text{pre}})''(v) \leq 0, \forall v \geq 0$. The marginal gain in diversity decreases as accumulated utility grows. Diminishing return is a well-understood fundamental principle of economics [2].

A3 (Capped potential). $C_\delta^{\text{pre}}(0) = 0, \lim_{v \rightarrow \infty} C_\delta^{\text{pre}}(v) = 1$. Diversity is zero before any exposure happens, and has an upper ceiling of 1.

A4 (Memoryless gains). For each fixed $\delta > 0$ and all $v, w \geq 0$,

$$C_\delta^{\text{pre}}(v + w) = C_\delta^{\text{pre}}(v) + (1 - C_\delta^{\text{pre}}(v)) \psi_\delta(w),$$

where $\psi_\delta(w)$ does not depend on the history variable v . A fixed extra utility w converts the *remaining* potential into diversity by the *same factor*, regardless of history [7, 8].

A5 (Threshold-invariant normalization). There exists $C_0 \in (0, 1)$ such that $C_\delta^{\text{pre}}(\delta) = C_0$ for every $\delta > 0$. The constant C_0 is independent of the threshold value.

These axioms are imposed only on the unsaturated curve C_δ^{pre} . They are not claimed for the implemented saturated function C_δ^* globally.

Based on these axioms, we can characterize the pre-threshold coverage family.

Theorem 1 (Pre-threshold exponential family). *For each $\delta > 0$, let $C_\delta^{\text{pre}} : [0, \infty) \rightarrow [0, 1)$ be a continuous nondecreasing function satisfying $C_\delta^{\text{pre}}(0) = 0$ and the memoryless gains identity*

$$C_\delta^{\text{pre}}(v + w) = C_\delta^{\text{pre}}(v) + (1 - C_\delta^{\text{pre}}(v)) \psi_\delta(w), \quad v, w \geq 0,$$

where ψ_δ is independent of the history variable v . Then $\psi_\delta(w) = C_\delta^{\text{pre}}(w)$ for all $w \geq 0$, and there exists a constant $k_\delta \geq 0$ such that for all $v \geq 0$,

$$C_\delta^{\text{pre}}(v) = 1 - e^{-k_\delta v}.$$

If A3 is also imposed, then $k_\delta > 0$; alternatively, the normalization condition below already implies $k_\delta > 0$. If the family-level normalization A5 additionally fixes $C_\delta^{\text{pre}}(\delta) = C_0$ for a constant $C_0 \in (0, 1)$ independent of δ , then

$$k_\delta = -\frac{\ln(1 - C_0)}{\delta}.$$

Choosing the normalization $C_0 = 1 - e^{-1}$ yields $k_\delta = 1/\delta$ and $C_\delta^{\text{pre}}(v) = 1 - e^{-v/\delta}$.

The diversity function used in our objective is the saturated extension

$$C_\delta^*(v) := C_\delta^{\text{pre}}(\min\{v, \delta\}) = \begin{cases} 1 - e^{-v/\delta}, & 0 \leq v < \delta, \\ 1 - e^{-1}, & \delta \leq v \leq 1, \end{cases}$$

which is a separate design choice consistent with the definition of aggregate coverage: after the threshold δ is crossed, additional utility no longer increases diversity. Theorem 1 characterizes the pre-threshold form; the saturation past δ is imposed externally and is not derived from A4.

It is easy to verify that the pre-threshold form $C_\delta^{\text{pre}}(v) = 1 - e^{-v/\delta}$ satisfies Axioms A1, A2, A4, and A5 under the chosen normalization. A3 holds in the limit $v \rightarrow \infty$ for the unsaturated form C_δ^{pre} . The implemented saturated function C_δ^* intentionally does not satisfy A3 or A4 globally, because after an item crosses the threshold its diversity contribution is fixed at $1 - e^{-1}$ by design.

Proof. Fix an arbitrary threshold $\delta > 0$. Setting $v = 0$ in A4 gives

$$C_\delta^{\text{pre}}(w) = C_\delta^{\text{pre}}(0) + (1 - C_\delta^{\text{pre}}(0))\psi_\delta(w) = \psi_\delta(w),$$

so the memoryless-gains identity can be written as

$$C_\delta^{\text{pre}}(v + w) = C_\delta^{\text{pre}}(v) + (1 - C_\delta^{\text{pre}}(v))C_\delta^{\text{pre}}(w).$$

Define the survival function

$$S_\delta(v) := 1 - C_\delta^{\text{pre}}(v), \quad v \geq 0.$$

The identity rearranges to

$$S_\delta(v + w) = (1 - C_\delta^{\text{pre}}(v))(1 - C_\delta^{\text{pre}}(w)) = S_\delta(v) S_\delta(w).$$

Thus S_δ satisfies Cauchy's multiplicative functional equation on $[0, \infty)$ with $S_\delta(0) = 1$ and S_δ continuous. The only continuous solutions of $S_\delta(v + w) = S_\delta(v)S_\delta(w)$ with $S_\delta(0) = 1$ are exponentials: there exists $k_\delta \in \mathbb{R}$ such that $S_\delta(v) = e^{-k_\delta v}$. Since C_δ^{pre} is nondecreasing, S_δ is nonincreasing, hence $k_\delta \geq 0$. If A3 is imposed, $S_\delta(v) \rightarrow 0$ as $v \rightarrow \infty$, which excludes $k_\delta = 0$ and gives $k_\delta > 0$. Therefore

$$C_\delta^{\text{pre}}(v) = 1 - e^{-k_\delta v}, \quad v \geq 0.$$

Imposing A5, $C_\delta^{\text{pre}}(\delta) = C_0$ gives $1 - e^{-k_\delta \delta} = C_0$, i.e. $k_\delta \delta = -\ln(1 - C_0)$. Selecting the normalization $C_0 = 1 - e^{-1}$ yields $k_\delta = 1/\delta$ and the pre-threshold form $C_\delta^{\text{pre}}(v) = 1 - e^{-v/\delta}$.

Finally, the saturated extension $C_\delta^*(v) = C_\delta^{\text{pre}}(\min\{v, \delta\})$ is adopted as a modeling choice to reflect the aggregate-coverage interpretation. This extension is not characterized by Axiom A4 beyond the threshold. $\square$

Theorem 1 shows that the pre-threshold family $C_\delta^{\text{pre}}(v) = 1 - e^{-v/\delta}$ is the unique continuous memoryless coverage family after the normalization $C_0 = 1 - e^{-1}$ is selected. A2 is then satisfied by this exponential form because its second derivative is nonpositive. The saturated extension is imposed as a modeling choice to encode aggregate coverage beyond the threshold and is the diversity function C_δ^* used throughout the paper.

1.2 Fairness Objective

In this section, we introduce how we design the fairness function based on the iso-elastic utility function and the Rawlsian principle in redistribution economics.

We design a function $U_\alpha^*(v)$ to pursue fairness in the second stage of recommendation when item utility v exceeds coverage threshold δ . The construction is inspired by Rawlsian redistribution and iso-elastic welfare [6, 4], but it should be understood mathematically as a separable concave welfare objective rather than as a literal conservation-law tax-and-transfer mechanism. The fairness function is:

$$U_\alpha^*(v) = \begin{cases} U_\alpha(\delta), & 0 \leq v < \delta, \\ U_\alpha(v), & \delta \leq v \leq 1. \end{cases}, \quad U_\alpha(v) = \begin{cases} \frac{\gamma}{1-\alpha} v^{1-\alpha}, & \alpha \neq 1, \\ \gamma \ln v, & \alpha = 1. \end{cases},$$

where $\alpha \in [0, \infty)$ denotes the fairness-aversion parameter, where larger values correspond to stronger marginal priority for lower post-threshold utilities. Throughout the theoretical analysis in this appendix, $\gamma > 0$ is a common scalar weight shared by all items; in experiments it is commonly set to 1. This scalar convention is the one used in Theorem 2 and Algorithm 1. A heterogeneous-weight extension with item weights γ_i is possible by defining item-specific functions $\mathcal{F}_i$ and item-specific smoothing coefficients $\lambda_i = \gamma_i e^{\delta^{1-\alpha}}$, in which case the smoothness and strong-concavity constants use $\gamma_{\max}$ and $\gamma_{\min}$, respectively.

The fairness component $U_\alpha(v)$ is increasing on the post-threshold domain, with marginal weight $U'_\alpha(v) = \gamma v^{-\alpha}$. It therefore does not impose an absolute penalty on high-utility items; rather, for $\alpha > 0$ it assigns smaller marginal reward to already high-utility items and larger marginal reward to lower-utility items. Under the scalar- γ convention, preserving the social welfare $\sum_i U_\alpha^*(v_i)$ implies that, at the margin, a unit gain for a lower-utility item j must be offset by a reduction from a higher-utility item i proportional to $(v_i/v_j)^\alpha$. Thus larger α creates stronger diminishing marginal returns

for high-utility items. For $\alpha > 1$, the unshifted iso-elastic value $\gamma v^{1-\alpha}/(1-\alpha)$ is negative on $(0, 1]$; the optimization uses its marginal structure, and affine shifts would not change argmax decisions.

When $\alpha = 0$, the post-threshold fairness component becomes linear in utility, matching the utilitarian branch of the iso-elastic family. The full DivFair objective still contains the thresholded diversity component and the constant pre-threshold fairness branch, so it should not be read as reducing globally to ordinary utilitarian top- K ranking. Setting $\alpha = 1$ gives a logarithmic post-threshold utility, corresponding to proportional fairness or Nash social welfare under the scalar-weight convention. It does not generally equalize item utilities unless the feasible set is symmetric and the items are exchangeable. As α increases, the marginal weight $\gamma v^{-\alpha}$ assigned to lower post-threshold utilities becomes larger relative to the weight assigned to higher utilities. This is related to the standard max-min limit of α -fair utility families, but in the present thresholded objective we use it only as an interpretation of increasing priority for lower-utility items rather than as a separate max-min convergence theorem.

1.3 Properties of Objective Function

After introducing the fairness and diversity objectives, we analyze the properties of the overall objective function (shown in Equation 1).

In redistribution economics, smoothness of economic transition is crucial when it comes to the turning point where the leading principle switches from Sufficientarian to Rawlsian. Likewise, to make our piecewise objective function smooth, it is essential to ensure a continuous function value and derivative value between two stages.

To ensure the smoothness of our objective function, we have the following theorem:

Theorem 2 (Properties of Function f). *Assume $\delta \in (0, 1]$, $\gamma > 0$, and $\alpha \geq 0$, and choose*

$$\lambda = \gamma e^{\delta^{1-\alpha}}$$

so that the derivative-matching condition $\lambda (C_\delta^)'(\delta^-) = (U_\alpha^*)'(\delta^+)$ holds (continuity of $\mathcal{F}'$ at the kink $v = \delta$). Then the objective function f defined in (1)–(2) is L -smooth on $[0, 1]^I$ with*

$$L = \gamma \delta^{-(\alpha+1)} \max\{e, \alpha\}.$$

Moreover, if $\alpha > 0$, then f is μ -strongly concave with

$$\mu = \gamma \min\{\alpha, \delta^{-(\alpha+1)}\} > 0.$$

For $\alpha = 0$ and $\delta < 1$, the post-threshold component is linear on a nontrivial interval, so f is concave (and L -smooth) but not strongly concave; any statement that explicitly invokes strong concavity therefore assumes $\alpha > 0$. The degenerate endpoint case $\alpha = 0, \delta = 1$ may retain curvature from the pre-threshold branch, but we do not use this special case.

Proof. From definitions in Section 1, we derive the first and second derivatives of $C_\delta^*(\cdot)$ and $U_\alpha^*(\cdot)$.

$$\begin{aligned} C_\delta^{*'}(v) &= \begin{cases} \frac{e^{-v/\delta}}{\delta}, & 0 \leq v < \delta, \\ 0, & \delta < v \leq 1, \end{cases} \\ C_\delta^{*''}(v) &= \begin{cases} -\frac{e^{-v/\delta}}{\delta^2}, & 0 \leq v < \delta, \\ 0, & \delta < v \leq 1. \end{cases} \\ U_\alpha'(v) &= \begin{cases} \gamma v^{-\alpha}, & \alpha \neq 1, \\ \frac{\gamma}{v}, & \alpha = 1, \end{cases} \quad U_\alpha^{*'}(v) = \begin{cases} 0, & 0 \leq v < \delta, \\ U_\alpha'(v), & \delta < v \leq 1, \end{cases} \\ U_\alpha''(v) &= \begin{cases} -\alpha\gamma v^{-\alpha-1}, & \alpha \neq 1, \\ -\frac{\gamma}{v^2}, & \alpha = 1. \end{cases} \quad U_\alpha^{*''}(v) = \begin{cases} 0, & 0 \leq v < \delta, \\ U_\alpha''(v), & \delta < v \leq 1, \end{cases} \end{aligned}$$

Therefore, with $\mathcal{F}(v) = \lambda C_\delta^*(v) + U_\alpha^*(v)$, we have

$$\mathcal{F}'(v) = \begin{cases} \lambda \frac{e^{-v/\delta}}{\delta}, & 0 \leq v < \delta, \\ \gamma v^{-\alpha}, & \delta < v \leq 1, \end{cases} \quad \mathcal{F}'(\delta^-) = \mathcal{F}'(\delta^+) = \frac{\lambda e^{-1}}{\delta} = \gamma \delta^{-\alpha},$$

using $\lambda = \gamma e \delta^{1-\alpha}$. We define $\mathcal{F}'(\delta) := \gamma \delta^{-\alpha}$ by this common one-sided derivative. Hence $\mathcal{F}'$ is continuous at the single "kink" point $v = \delta$.

$$\mathcal{F}''(v) = \begin{cases} -\lambda \frac{e^{-v/\delta}}{\delta^2}, & 0 \leq v < \delta, \\ -\alpha \gamma v^{-\alpha-1}, & \delta < v \leq 1, \\ \text{undefined (kink)}, & v = \delta. \end{cases}$$

By deriving the maximum and minimum of $|\mathcal{F}''(v)|$ on each piece:

- On $0 \leq v < \delta$, $|\mathcal{F}''(v)| = \lambda e^{-v/\delta} / \delta^2$ is maximized at $v = 0$ ($= \lambda / \delta^2 = \gamma e \delta^{-(\alpha+1)}$) and minimized at $v = \delta^-$ ($= \lambda e^{-1} / \delta^2 = \gamma \delta^{-(\alpha+1)}$).
- On $\delta < v \leq 1$, $|\mathcal{F}''(v)| = \alpha \gamma v^{-\alpha-1}$ is maximized at $v = \delta^+$ ($= \alpha \gamma \delta^{-(\alpha+1)}$) and (for $\alpha \geq 1$) minimized at $v = 1$ ($= \alpha \gamma$); for $0 < \alpha < 1$ the minimum is at $v = 1$.

Combining the bounds on the two pieces gives

$$L \triangleq \max_v |\mathcal{F}''(v)| = \gamma \delta^{-(\alpha+1)} \max\{e, \alpha\}, \quad \mu \triangleq \min_v |\mathcal{F}''(v)| = \gamma \min\{\alpha, \delta^{-(\alpha+1)}\}.$$

Note $\mu > 0$ requires $\alpha > 0$; for $\alpha = 0$ the post-threshold piece has $\mathcal{F}'' \equiv 0$, so $\mu = 0$ and f is only concave, not strongly concave.

We next derive curvature bounds on the smooth pieces and then prove smoothness and strong concavity using one-dimensional line-segment arguments. All Hessian statements below are understood away from the threshold hyperplanes and almost everywhere on $[0, 1]^I$; this is sufficient because each coordinate derivative $\mathcal{F}'$ is continuous and piecewise absolutely continuous with only one breakpoint.

Curvature bounds. Because f is separable, $\nabla^2 f(v) = \text{diag}(\mathcal{F}''(v_1), \dots, \mathcal{F}''(v_I))$. For $v_i \neq \delta$ the explicit formulas above give

$$|\mathcal{F}''(v_i)| \leq L,$$

and

$$|\mathcal{F}''(v_i)| \geq \mu.$$

Hence, away from the union of threshold hyperplanes (i.e., at points v with $v_i \neq \delta$ for every coordinate i),

$$-L \mathbf{I} \preceq \nabla^2 f(v) \preceq -\mu \mathbf{I}, \quad (3)$$

Because $\mathcal{F}'$ is continuous at δ and absolutely continuous on each open piece, for any line segment $z(s) = w + s(v - w)$ the map $s \mapsto \nabla f(z(s))$ is absolutely continuous. Its derivative equals $\nabla^2 f(z(s))(v - w)$ for almost every s ; the finitely many threshold-crossing values of s have measure zero and do not affect the integral.

L-smoothness. For any $v, w \in [0, 1]^I$ let $z(s) = w + s(v - w)$. By the one-dimensional fundamental theorem of calculus applied to the absolutely continuous map $s \mapsto \nabla f(z(s))$,

$$\nabla f(v) - \nabla f(w) = \int_0^1 \nabla^2 f(z(s))(v - w) ds$$

where the integrand is evaluated only almost everywhere. Using $\|\nabla^2 f(\cdot)\|_2 \leq L$ a.e. from (3),

$$\|\nabla f(v) - \nabla f(w)\|_2 \leq \int_0^1 L \|v - w\|_2 ds = L \|v - w\|_2.$$

μ -strong concavity. Let $d = w - v$ and define $h(s) = f(v + sd)$ for $s \in [0, 1]$. Since $\mathcal{F}'$ is continuous and piecewise absolutely continuous, h' is absolutely continuous and the Hessian exists almost everywhere along the segment. The standard second-order integral identity gives

$$f(w) - f(v) - \nabla f(v)^\top (w - v) = h(1) - h(0) - h'(0) = \int_0^1 (1 - s) d^\top \nabla^2 f(v + sd) ds.$$

Because $\nabla^2 f \preceq -\mu I$ almost everywhere along the segment, the integrand is at most $-\mu \|d\|_2^2$. Therefore

$$f(w) - f(v) - \nabla f(v)^\top (w - v) \leq -\frac{\mu}{2} \|w - v\|_2^2,$$

which is equivalent to $f(w) \leq f(v) + \nabla f(v)^\top (w - v) - \frac{\mu}{2} \|w - v\|_2^2$.

Because $\mathcal{F}'$ is continuous at δ and piecewise absolutely continuous with the displayed bounds, the smoothness and strong-concavity inequalities hold on all of $[0, 1]^I$.

Therefore, we showed that the objective function f defined in Section 1 is L -smooth on $[0, 1]^I$; and, when $\alpha > 0$, μ -strongly concave. For $\alpha = 0$, f is concave but not strongly concave. $\square$

Thus $\mathcal{F}$ is continuously differentiable, f is L -smooth on $[0, 1]^I$, and f is μ -strongly concave when $\alpha > 0$.

2 Our Approach: DivFair

Building on the design of the objective function, we propose a re-ranking algorithm, DivFair, which aims to efficiently optimize the objective proposed in Section 1. To achieve this, we formulate our multi-objective optimization problem as a mathematical program, and then propose an algorithm called DivFair for online recommendation.

2.1 Optimization Objective

First, define the one-round top- K action set and the full-horizon sequence set by

$$\mathcal{X}_K := \{\mathbf{x} \in \{0, 1\}^I : \mathbf{1}^\top \mathbf{x} = K\}, \quad \mathcal{X}_{1:T} := (\mathcal{X}_K)^T.$$

Throughout the optimization and algorithmic analysis, we assume normalized scores

$$0 \leq w_{t,i} \leq 1, \quad t = 1, \dots, T, \quad i \in \mathcal{I},$$

so every trajectory satisfies $0 \leq v_{t,i} = T^{-1} \sum_{s \leq t} w_{s,i} x_{s,i} \leq t/T \leq 1$ and all evaluations of f , ∇f , and the Taylor segments remain inside the domain $[0, 1]^I$. We also assume the integer exposure-cap problem is feasible: $K \leq I$ and there exists at least one sequence $\mathbf{x}_{1:T} \in \mathcal{X}_{1:T}$ with $T^{-1} \sum_t \mathbf{x}_t \leq \boldsymbol{\tau}$.

The multi-objective re-ranking can then be formulated as an integer programming problem. The optimization objective is

$$\begin{aligned} \mathcal{W}_{\text{OPT}} &= \max_{\mathbf{x}_{1:T} \in \mathcal{X}_{1:T}} \sum_{t=1}^T f(\mathbf{v}_t), \\ \text{s.t.} \quad \mathbf{v}_t &= \frac{1}{T} \sum_{s=1}^t \mathbf{w}_s \odot \mathbf{x}_s, \quad t = 1, \dots, T, \\ \bar{\mathbf{x}} &= \frac{1}{T} \sum_{t=1}^T \mathbf{x}_t \leq \boldsymbol{\tau}, \\ \mathbf{x}_t &\in \mathcal{X}_K, \quad t = 1, \dots, T. \end{aligned} \tag{4}$$

where f is the objective function applied to the accumulated utility state $\mathbf{v}_t$ induced by the ranking decisions up to time t . $\bar{\mathbf{x}} \in \mathbb{R}^I$ is the average exposure vector for items across all T users, and $\boldsymbol{\tau} \in [0, 1]^I$ denotes the exposure upper limits.

More specifically, based on Equation 1, the diverse and fair re-ranking problem is expressed as

$$\begin{aligned}
\mathcal{W}_{\text{OPT}} &= \max_{\mathbf{x}_{1:T} \in \mathcal{X}_{1:T}} \sum_{t=1}^T f(\mathbf{v}_t), \\
\text{s.t. } \bar{\mathbf{x}} &= \frac{1}{T} \sum_{t=1}^T \mathbf{x}_t \leq \boldsymbol{\tau}, \\
f(\mathbf{v}_t) &= \sum_{i \in \mathcal{I}} \lambda C_{\delta}^*(v_{t,i}) + U_{\alpha}^*(v_{t,i}), \\
v_{t,i} &= \frac{1}{T} \sum_{t'=1}^t w_{t',i} x_{t',i}, \forall i \in \mathcal{I}, \\
\mathbf{x}_t &\in \mathcal{X}_K, \quad t = 1, \dots, T.
\end{aligned} \tag{5}$$

where $\mathbf{x}_t \in \{0, 1\}^I$ is the decision vector for user u_t . For each item i , $x_{t,i} = 1$ if it is added to the ranking list $L_K(u_t)$, and $x_{t,i} = 0$ otherwise. The constraint $\mathbf{x}_t \in \mathcal{X}_K$ ensures that each recommendation list has size K . The first constraint in Equation 5 gives the accumulated average exposure, and the normalized-score assumption above ensures that $v_{t,i}$ is the normalized accumulated utility of item i up to time t , in the range $[0, 1]$.

We observe that directly optimizing Equation 5 is computationally challenging because it is a non-linear, large-scale integer program [3]. Although the objective is written as a sum over time, the cumulative state $\mathbf{v}_t$ couples decisions across rounds. This motivates an online algorithm that makes recommendations to user u_t without access to information beyond time step t . Next we will discuss our proposed model, DivFair, for online recommendation.

2.2 DivFair for Online Recommendation

In this section, we first derive a Lagrangian dual that upper-bounds the integer primal by a convex function of the dual variable. We then propose a projected online price-update rule, analyzed as projected gradient descent on auxiliary linear losses, for efficient online learning with provable error bounds.

Price clipping box. For the offline primal in Equation 5, we use a closed bounded price-comparator box only for the projected online-update analysis. Choose an arbitrary finite clipping radius $B > 0$ and define

$$\mathcal{U}_B := [0, B]^I.$$

No claim is made that this box contains an exact optimal multiplier for the integer offline problem, nor is any Slater-style multiplier-radius bound needed for the theorem below. The constants in the online-compatible comparator bound scale linearly with the chosen clipping radius B . The radius should therefore be treated as an algorithmic tuning parameter or a diagnostic clipping choice, not as a proved optimal dual radius for the total-welfare integer program.

Clipped weak-duality upper bound. From Equation 5 we know that directly solving this integer program offline is computationally challenging. For any clipped nonnegative price vector, the Lagrangian expression below provides a weak-duality upper bound on the primal optimum. Because $\mathbf{x}_t$ is integer and the sequence-level maximization remains state-coupled, this bound is motivational for clipped prices rather than a tractable exact dual reformulation or a derivation of the online update.

Theorem 3 (Clipped weak-duality upper bound). *For any nonnegative dual variable $\boldsymbol{\mu} \in \mathcal{U}_B = [0, B]^I$, the offline primal in Equation 4 satisfies*

$$\mathcal{W}_{\text{OPT}} \leq \mathcal{W}_{\text{clip}}(B) := \min_{\boldsymbol{\mu} \in \mathcal{U}_B} \left\{ \boldsymbol{\mu}^\top \boldsymbol{\tau} + \max_{\mathbf{x}_{1:T} \in \mathcal{X}_{1:T}} \left[\sum_{t=1}^T f(\mathbf{v}_t) - \boldsymbol{\mu}^\top \frac{1}{T} \sum_{t=1}^T \mathbf{x}_t \right] \right\}. \tag{6}$$

The clipped right-hand side $\mathcal{W}_{\text{clip}}(B)$ is a convex upper bound in the dual variable $\boldsymbol{\mu}$ because it is the pointwise maximum of affine functions of $\boldsymbol{\mu}$ plus a linear term. This is a weak-duality upper bound; it is not an equivalent convex reformulation of the integer primal.

Proof. Take any feasible sequence $\mathbf{x}_{1:T}$ of Equation 4. Its average exposure satisfies $T^{-1} \sum_{t=1}^T \mathbf{x}_t \leq \boldsymbol{\tau}$. Hence, for every $\boldsymbol{\mu} \in \mathcal{U}_B \subseteq \mathbb{R}_+^I$,

$$\sum_{t=1}^T f(\mathbf{v}_t) \leq \sum_{t=1}^T f(\mathbf{v}_t) + \boldsymbol{\mu}^\top \left(\boldsymbol{\tau} - \frac{1}{T} \sum_{t=1}^T \mathbf{x}_t \right).$$

Maximizing the left-hand side over all primal-feasible sequences and the right-hand side over the sequence set $\mathcal{X}_{1:T}$ gives, for every fixed $\boldsymbol{\mu} \in \mathcal{U}_B$,

$$\mathcal{W}_{\text{OPT}} \leq \boldsymbol{\mu}^\top \boldsymbol{\tau} + \max_{\mathbf{x}_{1:T} \in \mathcal{X}_{1:T}} \left[\sum_{t=1}^T f(\mathbf{v}_t) - \boldsymbol{\mu}^\top \frac{1}{T} \sum_{t=1}^T \mathbf{x}_t \right].$$

Taking the minimum over $\boldsymbol{\mu} \in \mathcal{U}_B$ proves (6). Since B is arbitrary, $\mathcal{W}_{\text{clip}}(B)$ should be read as a clipped Lagrangian upper bound, not as a true dual optimum. No auxiliary exposure variable is needed; the cap constraint is used directly in the Lagrangian inequality. $\square$

Since item utility $v_t = v_{t-1} + \frac{1}{T} w_t \odot x_t$ and $f(\cdot)$ depends on v_t , we use notation $f(v_t)$ in subsequent sections for brevity.

For per-round decisions, given the current exposure prefix $\boldsymbol{\theta}_t$, define the no-overflow masked action set

$$\mathcal{X}_t(\boldsymbol{\theta}_t) := \{\mathbf{x} \in \mathcal{X}_K : x_i = 0 \text{ whenever } \theta_{t,i} + 1/T > \tau_i\}.$$

Algorithm 1 uses this restricted set rather than an infinite penalty vector. The no-overflow theoretical version is a partial algorithm unless $\mathcal{X}_t(\boldsymbol{\theta}_t) \neq \emptyset$ at every round, equivalently unless at least K items remain unmasked at each realized round. A necessary finite-horizon capacity condition is

$$\sum_{i \in \mathcal{I}} \lfloor T\tau_i \rfloor \geq KT,$$

because item i can be selected at most $\lfloor T\tau_i \rfloor$ times without exceeding its average-exposure cap. This aggregate condition is not by itself sufficient for every online path, so the theorem below explicitly assumes nonemptiness of the realized masked sets. Thus there are two versions of the method: the *no-overflow theoretical version*, analyzed in Theorem 5, requires $\mathcal{X}_t(\boldsymbol{\theta}_t) \neq \emptyset$ for all t (equivalently, $|\{i : \theta_{t,i} + 1/T \leq \tau_i\}| \geq K$); the *practical fallback version*, used only when this set has size smaller than K , can select from $\mathcal{X}_K$ with an added overflow penalty such as $M \sum_i x_i (\theta_{t,i} + 1/T - \tau_i)_+$ and reports cap violation separately. The online-compatible comparator theorem below does not claim a bound for the fallback overflow loss unless that slack penalty is analyzed separately. Consequently, any experiment that invokes Theorem 5 as an explanation for a run should either have FallbackRate = 0 or report fallback frequency and cap-violation statistics separately.

DivFair algorithm. Algorithm 1 illustrates DivFair, our proposed method for online optimization. The hard mask enforces no-overflow feasibility when nonempty; the dual prices shape the ranking scores to discourage repeated selection of scarce items. DivFair leverages a linear approximation for real-time decision-making, and the projected update keeps the prices in a bounded box for stable online updates.

DivFair operates on sequences of users of length T . For each user sequence, the algorithm maintains three key components: a dual price vector $\boldsymbol{\mu}_t$, a phased budget $\boldsymbol{\theta}_t$ tracking the limits on average exposure, and the accumulated item utility vector $\mathbf{v}_t$. The procedure begins with initialization of $\boldsymbol{\mu}_1 = 0$, $\boldsymbol{\theta}_1 = 0$, and gradient $\mathbf{g}_0 = 0$. For each arriving user u_t , the algorithm forms the masked feasible set $\mathcal{X}_t(\boldsymbol{\theta}_t)$ to ensure that selected items do not violate the next-step exposure cap in the no-overflow regime. Lines 6–9 define the ideal one-step Lagrangian only as motivation; the implemented action is the first-order top- K surrogate $\mathbf{x}_t = \arg \max_{\mathbf{x} \in \mathcal{X}_t(\boldsymbol{\theta}_t)} [\nabla f(\mathbf{v}_{t-1})^\top (\mathbf{w}_t \odot \mathbf{x}) / T - \boldsymbol{\mu}_t^\top \mathbf{x} / T]$. Theorem 4 quantifies the difference between this surrogate and the exact one-step myopic Lagrangian.

Through this approximation, the scoring function becomes linear with respect to $\mathbf{x}_t$. Therefore, $\mathbf{x}_t$ can be computed using a top- K sorting algorithm with $\mathcal{O}(|\mathcal{I}| \log K)$ complexity using a heap (or linear-time selection variants), making it suitable for large-scale online recommendation scenarios.

Following the recommendation, we update the phased budget $\boldsymbol{\theta}_t$ and the utility vector $\mathbf{v}_t$, and set the auxiliary cap-side vector $\mathbf{a}_t = \boldsymbol{\tau}$. Then the per-round slack residual used for the dual-price update

Algorithm 1: DivFair

Input: User arriving order $\{u_i\}_{i=1}^T$, user size T , ranking size K , user-item preference score $w_{t,i}, \forall t, i$, and exposure constraints τ .
Output: The decision variables $\{\mathbf{x}_t\}_{t=1}^T$

- 1: Initialize dual solution $\boldsymbol{\mu}_1 = 0$, phased budget $\boldsymbol{\theta}_1 = 0$, item utility $\mathbf{v}_0 = 0$, and gradient $\mathbf{g}_0 = 0$.
- 2: **for** $t = 1, \dots, T$ **do**
- 3: User u_t arrives.
- 4: $\mathcal{X}_t(\boldsymbol{\theta}_t) = \{\mathbf{x} \in \mathcal{X}_K : x_i = 0 \text{ if } \theta_{t,i} + 1/T > \tau_i\}$ // anticipates the next-round exposure increment.
- 5: // Online recommendation
- 6: Define the ideal one-step objective $\mathcal{L}_t(\mathbf{x}) = f(\mathbf{v}_{t-1} + \mathbf{w}_t \odot \mathbf{x}/T) - \boldsymbol{\mu}_t^\top \mathbf{x}/T$.
- 7: Use its first-order surrogate $\tilde{\mathcal{L}}_t(\mathbf{x}) = f(\mathbf{v}_{t-1}) + \nabla f(\mathbf{v}_{t-1})^\top (\mathbf{w}_t \odot \mathbf{x})/T - \boldsymbol{\mu}_t^\top \mathbf{x}/T$.
- 8: Choose $\mathbf{x}_t = \arg \max_{\mathbf{x} \in \mathcal{X}_t(\boldsymbol{\theta}_t)} \tilde{\mathcal{L}}_t(\mathbf{x})$.
- 9: Equivalently, $\mathbf{x}_t = \arg \max_{\mathbf{x} \in \mathcal{X}_t(\boldsymbol{\theta}_t)} [(\nabla f(\mathbf{v}_{t-1}) \odot \mathbf{w}_t - \boldsymbol{\mu}_t)^\top \mathbf{x}/T]$.
- 10: $\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t + \mathbf{x}_t/T$
- 11: Choose $\mathbf{a}_t \in \arg \max_{\mathbf{a} \in [0, \tau]} \boldsymbol{\mu}_t^\top \mathbf{a}$, and set $\mathbf{a}_t = \tau$
- 12: $\mathbf{v}_t = \mathbf{v}_{t-1} + \mathbf{w}_t \odot \mathbf{x}_t/T$
- 13: $\mathbf{g}_t = (\mathbf{a}_t - \mathbf{x}_t)/T = (\tau - \mathbf{x}_t)/T$ // per-round slack residual used for the dual-price update; its sum equals the final exposure slack $\tau - \bar{\mathbf{x}}^o$ (see Section 2.3).
- 14: $\boldsymbol{\mu}_{t+1} = \arg \min_{\boldsymbol{\mu} \in \mathcal{U}_B} [\langle \mathbf{g}_t, \boldsymbol{\mu} \rangle + \|\boldsymbol{\mu} - \boldsymbol{\mu}_t\|^2/(2\eta)] = \Pi_{\mathcal{U}_B}(\boldsymbol{\mu}_t - \eta \mathbf{g}_t)$, where $\mathcal{U}_B := [0, B]^I$ is a bounded box (see Section 2.3).
- 15: **end for**

is $\mathbf{g}_t = (\mathbf{a}_t - \mathbf{x}_t)/T = (\tau - \mathbf{x}_t)/T$. The $1/T$ factor is the per-round share of the full exposure slack $\tau - \bar{\mathbf{x}}^o$: summing $\mathbf{g}_t$ across T rounds gives $\sum_t \mathbf{g}_t = \tau - (1/T) \sum_t \mathbf{x}_t^o = \tau - \bar{\mathbf{x}}^o$. In the regret proof, the update is analyzed as projected gradient descent on the linear analysis losses $\ell_t(\boldsymbol{\mu}) = \boldsymbol{\mu}^\top \mathbf{g}_t$; we do not claim that $\mathbf{g}_t$ is an exact subgradient of the full offline dual function, whose maximization is sequence-level and state-coupled. For a recommended item i ($x_{t,i} = 1$), $g_{t,i} = (\tau_i - 1)/T < 0$ (typical $\tau_i < 1$), so μ_i *increases* (item becomes more penalized in subsequent rounds); for an unrecommended item, $g_{t,i} = \tau_i/T > 0$, so μ_i *decreases* (item becomes less penalized). Intuitively, in classical welfare economics, the dual variable $\boldsymbol{\mu}$ can be viewed as the shadow price [6], reflecting the cost of allocating additional exposure to an item.

2.3 Theoretical Analysis

This section states the guarantees with their intended scope. We first bound the one-step linearization error, then prove a deterministic online-compatible comparator bound for DivFair. Regret to the fully clairvoyant offline optimum is reported through explicit compatibility-gap and drift diagnostics. Finally, we give a conditional P-MMF comparison template and stochastic-relaxation diagnostics. No result below should be read as an unconditional adversarial-regret theorem against the fully offline optimum or as a universal dominance theorem over P-MMF.

Scope of the theoretical claims.

Claim	Status	Location	Limitation
Smooth concave objective	Proven	Theorem 2	Requires the utility state to remain in $[0, 1]^I$.
One-step linearization error	Proven	Theorem 4	Myopic one-step Lagrangian approximation only.
Deterministic DivFair bound	Proven	Theorem 5	Relative to the realized path-dependent online-compatible comparator class.
Regret to $\mathcal{W}_{\text{OPT}}$	Conditional diagonal	Corollary 1	Requires small $\Gamma_T^\circ + \mathfrak{D}_{T,+}(\mathbf{y}^c)$.
P-MMF comparison	Conditional template	Proposition 1 and Corollary 2	Requires matched units, benchmark, and externally instantiated P-MMF constants.
Variance/concentration comparison	Diagnostic	Theorems 6–7	Applies to finite-temperature stochastic relaxations, not deterministic top- K variance.

Asymptotic convention. Unless explicitly stated otherwise, all big- O and rate statements treat $\alpha, \delta, \gamma, K, w_{\max}, B$, and the item dimension I as fixed constants. When these quantities vary with T , the bounds below should be read with their displayed constant dependence, for example through $L = \gamma\delta^{-(\alpha+1)} \max\{e, \alpha\}$.

Linear approximation. Benefiting from the properties of function f shown in Theorem 2, we have the following theorems to bound the error of linear approximation in Algorithm 1.

Lemma 1 (Taylor Expansion). *For any top- K vector $\mathbf{x}_t \in \mathcal{X}_K$ with $0 \leq w_{t,i} \leq w_{\max}$ and with the segment $\mathbf{v}_{t-1} + s(\mathbf{w}_t \odot \mathbf{x}_t/T)$ contained in $[0, 1]^I$ for all $s \in [0, 1]$ (in particular, under the global normalization $0 \leq w_{t,i} \leq 1$), the first-order Taylor approximation of f satisfies*

$$0 \leq \tilde{f}(v_t) - f(v_t) \leq \frac{Lw_{\max}^2 K}{2T^2}.$$

In particular, the approximation error is $\mathcal{O}(T^{-2})$ for fixed $L, w_{\max}, K$.

Proof. Let $\Delta_f(v_t) := \tilde{f}(v_t) - f(v_t)$ where $\tilde{f}(v_t) := f(v_{t-1}) + \nabla f(v_{t-1})^\top (v_t - v_{t-1})$ is the first-order Taylor expansion of f at v_{t-1} .

From the proof of Theorem 2 we know that $f : [0, 1]^I \rightarrow \mathbb{R}$ is L -smooth, with operator-norm Hessian bound $\|\nabla^2 f\|_2 \leq L$ almost everywhere on $[0, 1]^I$. Concavity of f gives $\Delta_f(v_t) \geq 0$.

Let $z_t := v_t - v_{t-1} = \frac{1}{T} \mathbf{w}_t \odot \mathbf{x}_t \in \mathbb{R}^I$. Parameterize the line segment between v_{t-1} and v_t by $\phi(s) := v_{t-1} + s z_t$ for $s \in [0, 1]$. By the multivariate Taylor remainder in integral form,

$$f(v_{t-1} + z_t) = f(v_{t-1}) + \nabla f(v_{t-1})^\top z_t + \int_0^1 (1-s) z_t^\top \nabla^2 f(v_{t-1} + s z_t) z_t ds.$$

Therefore

$$\begin{aligned} \Delta_f(v_t) &= \left| \tilde{f}(v_t) - f(v_t) \right| = \left| \int_0^1 (1-s) z_t^\top \nabla^2 f(v_{t-1} + s z_t) z_t ds \right| \\ &\leq \int_0^1 (1-s) \|\nabla^2 f(v_{t-1} + s z_t)\|_2 \|z_t\|_2^2 ds \leq \frac{L}{2} \|z_t\|_2^2 \\ &= \frac{L}{2T^2} \|\mathbf{w}_t \odot \mathbf{x}_t\|_2^2 \leq \frac{Lw_{\max}^2 K}{2T^2} \in \mathcal{O}(T^{-2}), \end{aligned}$$

where the last bound uses $\sum_i x_{t,i} = K$ and $|w_{t,i}| \leq w_{\max}$. $\square$

Using Lemma 1, we have Theorem 4 to bound the error of linear approximation of per-round Lagrangian $\mathcal{L}(\cdot)$.

Theorem 4 (Linear Approximation). *Let $\mathcal{L}_t(x) := f(\mathbf{v}_{t-1} + \mathbf{w}_t \odot x/T) - \boldsymbol{\mu}_t^\top x/T$ and $\tilde{\mathcal{L}}_t(x) := f(\mathbf{v}_{t-1}) + \nabla f(\mathbf{v}_{t-1})^\top (\mathbf{w}_t \odot x)/T - \boldsymbol{\mu}_t^\top x/T$ over a fixed masked set. Then, for every top- K feasible action x ,*

$$|\mathcal{L}_t(x) - \tilde{\mathcal{L}}_t(x)| \leq \frac{Lw_{\max}^2 K}{2T^2}.$$

If $x_t^ = \arg \max_x \mathcal{L}_t(x)$ and $x_t = \arg \max_x \tilde{\mathcal{L}}_t(x)$ over the same masked set, then*

$$0 \leq \mathcal{L}_t(x_t^*) - \mathcal{L}_t(x_t) \leq \frac{Lw_{\max}^2 K}{T^2}.$$

Proof. Let

$$\begin{aligned} \mathcal{L}_t(x_t) &:= f(\mathbf{v}_t) - \boldsymbol{\mu}_t^\top \mathbf{x}_t/T, \\ \tilde{\mathcal{L}}_t(x_t) &:= f(\mathbf{v}_{t-1}) + \nabla f(\mathbf{v}_{t-1})^\top (\mathbf{v}_t - \mathbf{v}_{t-1}) - \boldsymbol{\mu}_t^\top \mathbf{x}_t/T, \end{aligned}$$

and

$$\begin{aligned} \Delta_{\mathcal{L}}(x_t) &:= \mathcal{L}_t(x_t) - \tilde{\mathcal{L}}_t(x_t), \\ \epsilon_t &:= \mathcal{L}_t(x_t^*) - \mathcal{L}_t(x_t), \end{aligned}$$

where $x_t^* = \arg \max_{x \in \mathcal{X}_t(\theta_t)} \mathcal{L}_t(x)$ and $x_t = \arg \max_{x \in \mathcal{X}_t(\theta_t)} \tilde{\mathcal{L}}_t(x)$,

Let $v_t^* = v_{t-1} + \frac{1}{T} w_t \odot x_t^*$, $v_t = v_{t-1} + \frac{1}{T} w_t \odot x_t$. Using Lemma 1, $|\Delta_{\mathcal{L}}(x_t^*)| = |\Delta_f(v_t^*)| \in \mathcal{O}(T^{-2})$ and $|\Delta_{\mathcal{L}}(x_t)| = |\Delta_f(v_t)| \in \mathcal{O}(T^{-2})$ (these are the linearization gaps, equal up to additive terms independent of x).

By the definition $\Delta_{\mathcal{L}}(x) := \mathcal{L}_t(x) - \tilde{\mathcal{L}}_t(x)$, we have $\mathcal{L}_t(x) = \tilde{\mathcal{L}}_t(x) + \Delta_{\mathcal{L}}(x)$. Since $x_t = \arg \max_x \tilde{\mathcal{L}}_t(x)$,

$$\begin{aligned} \mathcal{L}_t(x_t^*) &= \tilde{\mathcal{L}}_t(x_t^*) + \Delta_{\mathcal{L}}(x_t^*) \\ &\leq \tilde{\mathcal{L}}_t(x_t) + \Delta_{\mathcal{L}}(x_t^*) \quad (\text{since } \tilde{\mathcal{L}}_t(x_t) \geq \tilde{\mathcal{L}}_t(x_t^*)) \\ &= \mathcal{L}_t(x_t) - \Delta_{\mathcal{L}}(x_t) + \Delta_{\mathcal{L}}(x_t^*). \end{aligned}$$

Hence

$$0 \leq \epsilon_t = \mathcal{L}_t(x_t^*) - \mathcal{L}_t(x_t) \leq \Delta_{\mathcal{L}}(x_t^*) - \Delta_{\mathcal{L}}(x_t) \leq |\Delta_{\mathcal{L}}(x_t^*)| + |\Delta_{\mathcal{L}}(x_t)|.$$

The explicit bound from Lemma 1 gives $|\Delta_{\mathcal{L}}(x)| \leq Lw_{\max}^2 K/(2T^2)$ for every feasible x , so $\epsilon_t \leq Lw_{\max}^2 K/T^2$. This completes the proof. $\square$

From Theorem 4, the linear approximation contributes at most $Lw_{\max}^2 K/T^2$ per round, hence at most $Lw_{\max}^2 K/T$ cumulatively over T rounds.

Online-compatible comparator bound and full-offline diagnostic decomposition. We measure the regret of an algorithm as the difference between the optimal performance $\mathcal{W}_{OPT}$ of the offline problem and the performance of the online algorithm.

$$\text{Regret}(f) = \mathcal{W}_{OPT} - \mathcal{W}_{\text{online}}. \quad (7)$$

Setup for Theorem 5. We now state the guarantee directly in terms of the original welfare objective, but with a benchmark that is compatible with the online hard mask actually used by Algorithm 1. Let

$$\mathcal{W}_{\text{online}} := \sum_{t=1}^T f(\mathbf{v}_t^o)$$

be the welfare obtained by Algorithm 1. Because the hard exposure mask depends on the online prefix, define the realized online masked action set

$$\mathcal{X}_t^o := \{\mathbf{x} \in \{0, 1\}^I : \sum_i x_i = K, x_i = 0 \text{ whenever } \theta_{t,i}^o + 1/T > \tau_i\}.$$

The realized path-dependent online-compatible comparator class is

$$\mathfrak{C}^o := \left\{ \mathbf{y}_{1:T} \in (\mathcal{X}_K)^T : \mathbf{y}_t \in \mathcal{X}_t^o \ \forall t, \quad \frac{1}{T} \sum_{t=1}^T \mathbf{y}_t \leq \boldsymbol{\tau} \right\}. \quad (8)$$

When the realized masks are nonempty, Algorithm 1's own trajectory belongs to this class, so $\mathfrak{C}^o \neq \emptyset$. For any comparator $\mathbf{y}_{1:T} \in \mathfrak{C}^o$, set

$$\mathbf{v}_t^{\mathbf{y}} := \frac{1}{T} \sum_{s \leq t} \mathbf{w}_s \odot \mathbf{y}_s, \quad \mathcal{W}(\mathbf{y}) := \sum_{t=1}^T f(\mathbf{v}_t^{\mathbf{y}}),$$

and define the data-dependent state-coupling drift

$$\mathfrak{D}_T(\mathbf{y}) := \sum_{t=1}^T \nabla f(\mathbf{v}_{t-1}^o)^\top (\mathbf{v}_{t-1}^{\mathbf{y}} - \mathbf{v}_{t-1}^o), \quad \mathfrak{D}_{T,+}(\mathbf{y}) := \max\{\mathfrak{D}_T(\mathbf{y}), 0\}. \quad (9)$$

Let

$$\mathbf{y}_{1:T}^c \in \arg \max_{\mathbf{y}_{1:T} \in \mathfrak{C}^o} \mathcal{W}(\mathbf{y}), \quad \mathcal{W}_{\text{comp}}^o := \mathcal{W}(\mathbf{y}^c).$$

Finally, let $\mathbf{x}_{1:T}^*$ be an optimal solution of the original offline problem in Equation 5, and define

$$\mathcal{W}_{\text{OPT}} := \sum_{t=1}^T f(\mathbf{v}_t^*), \quad \Gamma_T^o := \mathcal{W}_{\text{OPT}} - \mathcal{W}_{\text{comp}}^o \geq 0.$$

The nonnegative quantity Γ_T^o is the *realized compatibility gap*: it is zero exactly when at least one offline optimum also lies in the online-compatible class $\mathfrak{C}^o$. This formulation removes the earlier need to assume, inside the theorem, that the clairvoyant optimum is compatible with the algorithm's realized masks. It also does not assume a sublinear drift rate; the drift appears as an explicit, logged quantity in the bound. A sublinear regret rate against the fully clairvoyant offline optimum requires separate evidence that $\Gamma_T^o + \mathfrak{D}_{T,+}(\mathbf{y}^c)$ is sublinear, which can be checked post hoc or derived from a separate stochastic arrival model.

Deterministic drift diagnostics. For any $\mathbf{y}_{1:T} \in \mathfrak{C}^o$, let

$$M_{\nabla} := \sup_{\mathbf{v} \in [0,1]^I} \|\nabla f(\mathbf{v})\|_{\infty} \leq \gamma e \delta^{-\alpha},$$

where the bound follows from the derivative formulas in Theorem 2. Since

$$\mathbf{v}_{t-1}^{\mathbf{y}} - \mathbf{v}_{t-1}^o = \frac{1}{T} \sum_{s < t} \mathbf{w}_s \odot (\mathbf{y}_s - \mathbf{x}_s^o),$$

we have the deterministic inequality

$$\mathfrak{D}_T(\mathbf{y}) \leq \frac{M_{\nabla}}{T} \sum_{t=1}^T \left\| \sum_{s < t} \mathbf{w}_s \odot (\mathbf{y}_s - \mathbf{x}_s^o) \right\|_1. \quad (10)$$

Consequently, if the weighted prefix discrepancy of a chosen comparator is bounded by

$$\left\| \sum_{s < t} \mathbf{w}_s \odot (\mathbf{y}_s - \mathbf{x}_s^o) \right\|_1 \leq C_{\Delta}(\mathbf{y}) \sqrt{t}, \quad t = 1, \dots, T,$$

then

$$\mathfrak{D}_T(\mathbf{y}) \leq \frac{M_{\nabla} C_{\Delta}(\mathbf{y})}{T} \sum_{t=1}^T \sqrt{t} \leq \frac{2}{3} M_{\nabla} C_{\Delta}(\mathbf{y}) \sqrt{T} + M_{\nabla} C_{\Delta}(\mathbf{y}). \quad (11)$$

The prefix-discrepancy constant and the compatibility gap are not hidden assumptions in Theorem 5; they are observable diagnostics once the online trajectory and comparator have been computed. Random-order or i.i.d. arrivals with sufficient cap slack are natural regimes in which such prefix discrepancies may scale like $\sqrt{t}$, while an adversarial ordering can make them scale like t and hence can force linear regret to the fully clairvoyant offline optimum.

For the proof, define the online-anchor linearization

$$\tilde{f}_t(\mathbf{x}) := f(\mathbf{v}_{t-1}^\circ) + \frac{1}{T} \nabla f(\mathbf{v}_{t-1}^\circ)^\top (\mathbf{w}_t \odot \mathbf{x}),$$

and the corresponding Taylor remainder

$$r_t(\mathbf{x}) := \tilde{f}_t(\mathbf{x}) - f(\mathbf{v}_{t-1}^\circ + \mathbf{w}_t \odot \mathbf{x}/T).$$

By concavity and Lemma 1, for every top- K vector $\mathbf{x}$,

$$0 \leq r_t(\mathbf{x}) \leq \frac{Lw_{\max}^2 K}{2T^2}. \quad (12)$$

Finally, Algorithm 1 uses the averaged residual

$$\mathbf{g}_t = \frac{\boldsymbol{\tau} - \mathbf{x}_t^\circ}{T}, \quad \|\mathbf{g}_t\|_2 \leq \frac{\bar{G}}{T}, \quad \bar{G} := \|\boldsymbol{\tau}\|_2 + \sqrt{K}.$$

A decreasing learning rate such as $\eta = \Theta(T^{-1/2})$ is a stable online-update convention. The factor $1/T$ is already inside $\mathbf{g}_t$, and the bound below holds for every fixed $\eta > 0$.

Theorem 5 (Realized path-compatible comparator inequality for DivFair). *Assume $0 \leq w_{t,i} \leq 1$, $\boldsymbol{\tau} \in [0, 1]^I$, $\mathcal{X}_t^\circ \neq \emptyset$ for every t , and Algorithm 1 projects onto $\mathcal{U}_B = [0, B]^I$ for a finite clipping radius $B > 0$. Then, for every comparator $\mathbf{y}_{1:T} \in \mathfrak{C}^\circ$ and every learning rate $\eta > 0$,*

$$\mathcal{W}(\mathbf{y}) - \mathcal{W}_{\text{online}} \leq \mathfrak{D}_{T,+}(\mathbf{y}) + BK + \frac{\eta \bar{G}^2}{2T} + \frac{Lw_{\max}^2 K}{2T}. \quad (13)$$

In particular,

$$\mathcal{W}_{\text{comp}}^\circ - \mathcal{W}_{\text{online}} \leq \mathfrak{D}_{T,+}(\mathbf{y}^c) + BK + \frac{\eta \bar{G}^2}{2T} + \frac{Lw_{\max}^2 K}{2T}, \quad (14)$$

and the regret to the fully clairvoyant offline optimum satisfies the diagnostic decomposition

$$\mathcal{W}_{\text{OPT}} - \mathcal{W}_{\text{online}} \leq \Gamma_T^\circ + \mathfrak{D}_{T,+}(\mathbf{y}^c) + BK + \frac{\eta \bar{G}^2}{2T} + \frac{Lw_{\max}^2 K}{2T}. \quad (15)$$

Equivalently, the average online-compatible regret is obtained by dividing the right-hand side of (14) by T .

Proof. Fix any $\mathbf{y}_{1:T} \in \mathfrak{C}^\circ$. Because $\mathbf{y}_t \in \mathcal{X}_t^\circ$, and Algorithm 1 chooses

$$\mathbf{x}_t^\circ \in \arg \max_{\mathbf{x} \in \mathcal{X}_t^\circ} \left\{ \tilde{f}_t(\mathbf{x}) - \frac{1}{T} \boldsymbol{\mu}_t^\top \mathbf{x} \right\},$$

we have, for every t ,

$$\tilde{f}_t(\mathbf{y}_t) - \tilde{f}_t(\mathbf{x}_t^\circ) \leq \frac{1}{T} \boldsymbol{\mu}_t^\top (\mathbf{y}_t - \mathbf{x}_t^\circ) = \frac{1}{T} \boldsymbol{\mu}_t^\top (\mathbf{y}_t - \boldsymbol{\tau}) + \boldsymbol{\mu}_t^\top \mathbf{g}_t. \quad (16)$$

Since $0 \leq \mu_{t,i} \leq B$, $\boldsymbol{\tau} \geq 0$, and $\|\mathbf{y}_t\|_1 = K$,

$$\sum_{t=1}^T \frac{1}{T} \boldsymbol{\mu}_t^\top (\mathbf{y}_t - \boldsymbol{\tau}) \leq \sum_{t=1}^T \frac{1}{T} \boldsymbol{\mu}_t^\top \mathbf{y}_t \leq BK. \quad (17)$$

This constant is the price of comparing the hard-masked online policy with a comparator sequence through bounded prices; it is independent of T .

Step 2: projected gradient controls the residual term. Let $\ell_t(\boldsymbol{\mu}) := \boldsymbol{\mu}^\top \mathbf{g}_t$. The update

$$\boldsymbol{\mu}_{t+1} = \Pi_{\mathcal{U}_B}(\boldsymbol{\mu}_t - \eta \mathbf{g}_t)$$

is projected gradient descent on the linear analysis losses ℓ_t . For every fixed $\boldsymbol{\mu} \in \mathcal{U}_B$,

$$\sum_{t=1}^T \ell_t(\boldsymbol{\mu}_t) - \sum_{t=1}^T \ell_t(\boldsymbol{\mu}) \leq \frac{\|\boldsymbol{\mu} - \boldsymbol{\mu}_1\|_2^2}{2\eta} + \frac{\eta}{2} \sum_{t=1}^T \|\mathbf{g}_t\|_2^2.$$

Taking $\boldsymbol{\mu} = \mathbf{0}$, using $\boldsymbol{\mu}_1 = \mathbf{0}$, and using $\|\mathbf{g}_t\|_2 \leq \bar{G}/T$,

$$\sum_{t=1}^T \boldsymbol{\mu}_t^\top \mathbf{g}_t \leq \frac{\eta}{2} \sum_{t=1}^T \|\mathbf{g}_t\|_2^2 \leq \frac{\eta \bar{G}^2}{2T}. \quad (18)$$

Combining (16), (17), and (18) yields

$$\sum_{t=1}^T \tilde{f}_t(\mathbf{y}_t) - \sum_{t=1}^T \tilde{f}_t(\mathbf{x}_t^\circ) \leq BK + \frac{\eta \bar{G}^2}{2T}. \quad (19)$$

Step 3: convert the linearized comparison to true welfare. For the online trajectory, (12) gives

$$0 \leq \sum_{t=1}^T [\tilde{f}_t(\mathbf{x}_t^\circ) - f(\mathbf{v}_t^\circ)] \leq \frac{Lw_{\max}^2 K}{2T}. \quad (20)$$

For the comparator trajectory, concavity of f at the online state $\mathbf{v}_{t-1}^\circ$ implies

$$\begin{aligned} f(\mathbf{v}_t^\mathbf{y}) &\leq f(\mathbf{v}_{t-1}^\circ) + \nabla f(\mathbf{v}_{t-1}^\circ)^\top (\mathbf{v}_t^\mathbf{y} - \mathbf{v}_{t-1}^\circ) \\ &= \tilde{f}_t(\mathbf{y}_t) + \nabla f(\mathbf{v}_{t-1}^\circ)^\top (\mathbf{v}_{t-1}^\mathbf{y} - \mathbf{v}_{t-1}^\circ). \end{aligned}$$

Summing this inequality over t gives

$$\mathcal{W}(\mathbf{y}) \leq \sum_{t=1}^T \tilde{f}_t(\mathbf{y}_t) + \mathfrak{D}_T(\mathbf{y}). \quad (21)$$

Combining (19), (20), and (21), and using $\mathfrak{D}_T(\mathbf{y}) \leq \mathfrak{D}_{T,+}(\mathbf{y})$, proves (13). Applying (13) to $\mathbf{y}^c$ proves (14). Adding $\Gamma_T^\circ = \mathcal{W}_{\text{OPT}} - \mathcal{W}_{\text{comp}}^\circ$ gives (15). $\square$

Corollary 1 (Diagnostic rate consequence). *Under the assumptions of Theorem 5, if the compatible-comparator drift satisfies $\mathfrak{D}_{T,+}(\mathbf{y}^c) = \mathcal{O}(\sqrt{T})$, then*

$$\mathcal{W}_{\text{comp}}^\circ - \mathcal{W}_{\text{online}} = \mathcal{O}(\sqrt{T})$$

for fixed $\alpha, \delta, \gamma, I, B, K$ and any stable decreasing learning-rate convention such as $\eta = \eta_0/\sqrt{T}$. If additionally the realized compatibility gap satisfies $\Gamma_T^\circ = \mathcal{O}(\sqrt{T})$, then

$$\mathcal{W}_{\text{OPT}} - \mathcal{W}_{\text{online}} = \mathcal{O}(\sqrt{T}).$$

If the drift and compatibility gap are $\mathcal{O}(1)$, the corresponding bounds improve to $\mathcal{O}(1)$.

Remarks on the comparator theorem. Theorem 5 is a deterministic bound under the stated normalization, no-overflow, and clipping assumptions, relative to the realized path-dependent online-compatible comparator class $\mathfrak{C}^\circ$, plus the explicit diagnostic decomposition (15) for the fully clairvoyant offline optimum. It is not a distribution-free adversarial regret theorem against $\mathcal{W}_{\text{OPT}}$. Sublinear regret to $\mathcal{W}_{\text{OPT}}$ follows only after separate empirical or model-specific evidence shows that $\Gamma_T^\circ + \mathfrak{D}_{T,+}(\mathbf{y}^c)$ is sublinear. The auxiliary linearized quantities appear only inside the proof. The projected price update is analyzed as projected gradient descent on the linear analysis losses $\ell_t(\boldsymbol{\mu}) = \boldsymbol{\mu}^\top \mathbf{g}_t$, not as exact mirror descent on the sequence-level offline dual.

Conditional comparison protocol with P-MMF. We compare DivFair with the representative re-ranking model P-MMF of Xu et al. [9], adapted to individual-level item fairness in online recommendation. To avoid comparing unrelated constants, we fix the benchmark explicitly: both methods are evaluated on the same item universe, exposure caps, horizon, score normalization, and welfare benchmark. The P-MMF theorem we instantiate is the regret-bound theorem for P-MMF in Xu et al. [9], after translating provider resources to item-level exposure resources and using the same T -normalized exposure convention as in Equation 5.

Proposition 1 (Conditional bookkeeping protocol under an externally instantiated P-MMF coefficient). *Assume the P-MMF regret theorem is invoked under a matched item-level exposure benchmark and has the leading-coefficient form*

$$\mathcal{W}_{\text{OPT}} - \mathcal{W}_{\text{online}}^{\text{P}} \leq C_{\text{P}}^{\text{inst}} \sqrt{T} + o(\sqrt{T}), \quad C_{\text{P}}^{\text{inst}} := H_P + \frac{G_P^2}{c} \left(1 + \frac{\sigma}{2c}\right), \quad c = (1 - \rho_{\text{mom}})\sigma. \quad (22)$$

Here G_P is the P-MMF resource-gradient norm bound under the matched norm convention, $H_P \geq 0$ is the residual/regularizer constant appearing in the invoked P-MMF theorem, $\sigma > 0$ is the resource-imbalance or strong-convexity parameter used by that theorem, and $\rho_{\text{mom}} \in (0, 1)$ is P-MMF's momentum parameter. If a different P-MMF theorem or normalization is used, then $C_{\text{P}}^{\text{inst}}$ must be recomputed from that theorem rather than reused from (22). This proposition is a conditional bookkeeping template: it is not a self-contained proof that P-MMF has coefficient $C_{\text{P}}^{\text{inst}}$, and it is not a universal DivFair-superiority theorem unless the invoked P-MMF theorem and the item-level mapping are supplied and verified.

Under Theorem 5, DivFair's full-offline diagnostic leading term is controlled by the compatibility gap and compatible-comparator drift. If

$$\Gamma_T^{\circ} \leq C_{\Gamma} \sqrt{T} + o(\sqrt{T}), \quad \mathfrak{D}_{T,+}(\mathbf{y}^c) \leq C_{\text{drift}} \sqrt{T} + o(\sqrt{T}),$$

then Theorem 5 gives

$$\mathcal{W}_{\text{OPT}} - \mathcal{W}_{\text{online}}^{\text{D}} \leq (C_{\Gamma} + C_{\text{drift}}) \sqrt{T} + o(\sqrt{T}). \quad (23)$$

By (11), a logged prefix-discrepancy constant gives the computable sufficient estimate

$$C_{\text{drift}} \leq \frac{2}{3} M_{\nabla} C_{\Delta}(\mathbf{y}^c) + o(1) \quad \text{or, more conservatively,} \quad C_{\text{drift}} \lesssim M_{\nabla} C_{\Delta}(\mathbf{y}^c). \quad (24)$$

Therefore, after all benchmark and unit conventions have been aligned, a sufficient condition for DivFair to have a smaller leading coefficient than the instantiated P-MMF bound is

$$C_{\Gamma} + C_{\text{drift}} < C_{\text{P}}^{\text{inst}}. \quad (25)$$

In the special case $\Gamma_T^{\circ} = o(\sqrt{T})$, this reduces to $C_{\text{drift}} < C_{\text{P}}^{\text{inst}}$. If the compatibility gap is not sublinear or the constants are not mapped under the same benchmark, the theory supports only an empirical comparison.

Illustrative numerical regimes (scale sanity check only). The following calculation is a scale sanity check, not proof-level evidence of dominance. It is borrowed as an illustrative comparison and should be used only after the P-MMF theorem and the DivFair benchmark have been mapped as above. Take the instantiated P-MMF coefficient in (22). For a top- K selection vector, a typical scale is $G_P^2 \asymp K$ under the corresponding P-MMF norm convention. Taking $H_P = 0$ only makes the comparison harder for DivFair, since any positive H_P increases $C_{\text{P}}^{\text{inst}}$. Representative values are

K	$(\rho_{\text{mom}}, \sigma)$	c	$C_{\text{P}}^{\text{inst}}$
20	(0.60, 0.30)	0.12	$20/0.12 (1 + 0.30/0.24) \approx 375$,
20	(0.50, 0.40)	0.20	$20/0.20 (1 + 0.40/0.40) = 200$,
20	(0.70, 0.20)	0.06	$20/0.06 (1 + 0.20/0.12) \approx 889$.

On the DivFair side, $M_{\nabla} \leq \gamma e \delta^{-\alpha}$. For example, with $\gamma = 1$, $\delta = 0.1$, and $\alpha = 0.5$, we have $M_{\nabla} \leq e\sqrt{10} \approx 8.60$. If the logged prefix-discrepancy constant is $C_{\Delta}(\mathbf{y}^c) = 1, 3, 5$, then (11) gives

$$C_{\text{drift}} \lesssim (2/3) M_{\nabla} C_{\Delta} \in \{5.7, 17.2, 28.7\}.$$

In a more conservative curvature regime, say $\alpha = 1$ and $\delta = 0.1$, $M_{\nabla} \leq 10e \approx 27.2$; then $C_{\Delta} = 1, 3, 5$ gives

$$C_{\text{drift}} \lesssim \{18.1, 54.4, 90.6\}.$$

If $C_{\Gamma} = 0$, these values are below the illustrative P-MMF coefficient $C_{\text{P}}^{\text{inst}} = 200$. More generally, the comparison must include C_{Γ} : under the weakest displayed P-MMF coefficient, DivFair’s diagnostic bound is smaller whenever $C_{\Gamma} + (2/3)M_{\nabla}C_{\Delta} < 200$. Thus this section should be read as an instantiated, practically checkable comparison recipe, not as a universal statement for all adversarial sequences.

Corollary 2 (Matched leading-coefficient comparison with P-MMF). *Assume the matched P-MMF bound (22) holds and the DivFair diagnostics satisfy $\Gamma_T^{\circ} \leq C_{\Gamma}\sqrt{T} + o(\sqrt{T})$ and $\mathfrak{D}_{T,+}(\mathbf{y}^c) \leq C_{\text{drift}}\sqrt{T} + o(\sqrt{T})$. Then DivFair has a smaller leading $\sqrt{T}$ coefficient than this instantiated P-MMF bound whenever*

$$C_{\Gamma} + C_{\text{drift}} < C_{\text{P}}^{\text{inst}}.$$

If this inequality is not verified for the concrete instance and the chosen P-MMF parameterization, the theory supports only an empirical comparison, not a universal “strictly tighter” claim.

Proof. The P-MMF side is (22). The DivFair side follows from (15) and the two diagnostic rate assumptions:

$$\mathcal{W}_{\text{OPT}} - \mathcal{W}_{\text{online}}^{\text{D}} \leq (C_{\Gamma} + C_{\text{drift}})\sqrt{T} + BK + o(\sqrt{T}) = (C_{\Gamma} + C_{\text{drift}})\sqrt{T} + o(\sqrt{T}),$$

since BK is independent of T . Comparing the leading coefficients proves the result. The conclusion is conditional on the matched theorem and diagnostic constants, and should not be read as a universal superiority theorem. $\square$

In this section, we compare DivFair with a representative re-ranking model P-MMF, adapted to individual-level item fairness. We provide Theorem 6 to compare martingale-noise components of the projected native update variables, and Theorem 7 to compare the concentration of centered native-update noise under finite-temperature stochastic relaxations. These results are conditional stochastic-relaxation diagnostics; they do not constitute an unconditional convergence proof for the full deterministic dual trajectory.

Stochastic relaxation Algorithm 1-S and notations. The following variance and concentration results compare the stochastic-noise components of the native update variables used by each method. DivFair’s native exposure residual is horizon-normalized, $\mathbf{g}_t^{\text{D}} = (\boldsymbol{\tau} - \mathbf{x}_t)/T$, because the objective constrains average exposure. The P-MMF analogue below follows the published momentum recursion in its native instantaneous-resource units. Therefore, Theorems 6–7 are not same-unit deterministic convergence theorems. If P-MMF is rewritten with the same extra $1/T$ horizon normalization, its bounds should be rescaled by the corresponding $1/T$ factor and the order separation becomes a constant-level comparison.

The deterministic top- K **argmax** in line 6 of Algorithm 1 makes $\mathbf{x}_t$ a deterministic function of the score vector, so it has no sampling variance under the deployed policy. To carry out a meaningful variance/concentration analysis, we analyze the following *finite-temperature stochastic relaxation* of Algorithm 1, denoted *Algorithm 1-S*: all lines of Algorithm 1 are unchanged except line 6, which is replaced by Gumbel-top- K sampling at a fixed temperature $\beta > 0$,

$$\mathbf{x}_t = \text{argTopK}_{i \in A_t} \left(\text{score}_{t,i} + \frac{1}{\beta} G_{t,i} \right),$$

where $\{G_{t,i}\}$ are i.i.d. standard Gumbel random variables [5] and

$$A_t := \{i : \theta_{t,i} + 1/T \leq \tau_i\}$$

is the currently unmasked candidate set. Masked items are excluded from the sampling set, equivalently they have probability zero. We denote the P-MMF analogue, obtained by applying the same finite-temperature Gumbel-top- K exploration on top of P-MMF’s score, by analogous notation.

The user arrival u_t , the preference vector w_t , the score vector, and the mask A_t are treated as predictable before the round- t sampling randomness is drawn. Formally, we condition on the pre-sampling filtration

$$\mathcal{H}_t := \sigma(\mathbf{x}_1, \dots, \mathbf{x}_{t-1}, u_t, w_t, \text{score}_t, A_t),$$

and let $\mathcal{F}_t := \sigma(\mathcal{H}_t, \mathbf{x}_t)$ be the post-sampling filtration. All conditional probabilities and variances below are with respect to $\mathcal{H}_t$.

We use superscripts D and P to represent DivFair (Algorithm 1-S) and P-MMF, respectively. Let S_t be the size- K set recommended at round t . The marginal sampling probabilities

$$\pi_{t,i}^D := \Pr(i \in S_t \mid \mathcal{H}_t), \quad \pi_{t,i}^P := \Pr(i \in S_t \mid \mathcal{H}_t)$$

are $\mathcal{H}_t$ -measurable. For each fixed coordinate i , $x_{t,i} \mid \mathcal{H}_t$ is Bernoulli with marginal probability $\pi_{t,i}$. This marginal Bernoulli statement does not assert coordinate-wise independence: under fixed-size top- K sampling, the coordinates of $\mathbf{x}_t$ are generally dependent because exactly K items are selected.

To avoid conflict with the accumulated utility notation $v_{t,i}$ used earlier, define the conditional Bernoulli variance by

$$\sigma_{t,i}^{2,D} := \pi_{t,i}^D(1 - \pi_{t,i}^D), \quad \sigma_{t,i}^{2,P} := \pi_{t,i}^P(1 - \pi_{t,i}^P).$$

We write

$$\sigma_{\max}^{2,D/P} := \sup_{t,i} \sigma_{t,i}^{2,D/P} \leq \frac{1}{4}.$$

Theorem 6 only needs the DivFair upper bound above. If one needs a lower bound on sampling probabilities, it must be stated only over currently unmasked items and under the corresponding score-boundedness condition; no positive lower bound is claimed for masked items, since they have $\pi_{t,i} = 0$.

In P-MMF, $\gamma_{\max}$ denotes the maximum resource and $\rho_{\text{mom}} \in (0, 1)$ the momentum coefficient (renamed from P-MMF's α to avoid conflict with DivFair's fairness parameter). The quantities $\text{Var}[\mu_{T,i}^D]$ and $\text{Var}[\mu_{T,i}^{P,\perp}]$ denote the variances of the *martingale stochastic-noise parts* of the corresponding dual variables, in the sense of the Doob decompositions used in Lemma 2 and Theorem 6.

Item-level P-MMF stochastic analogue. For the diagnostic comparison only, let $\tilde{g}_t$ denote the primitive item-level P-MMF resource residual before momentum and projection. To make the analogue explicit, we write each coordinate in the form

$$\tilde{g}_{t,i} = b_{t,i}^P - r_{t,i}^P x_{t,i},$$

where $b_{t,i}^P$ and $r_{t,i}^P$ are $\mathcal{H}_t$ -measurable predictable quantities in the native P-MMF resource units. The scalar $r_{t,i}^P$ captures any item-resource weight, provider-to-item conversion factor, or unit convention. We assume the bounded-resource condition

$$|r_{t,i}^P| \leq R_P < \infty \quad \text{for all } t, i,$$

with $R_P = 1$ only in the special unit-indicator residual case. Conditional on the pre-sampling filtration $\mathcal{H}_t$, all deterministic score, resource, and projection terms are measurable, and the stochastic part comes only from the sampled finite-temperature top- K action. Therefore

$$\tilde{Y}_{t,i} := \tilde{g}_{t,i} - \mathbb{E}[\tilde{g}_{t,i} \mid \mathcal{H}_t] = -r_{t,i}^P(x_{t,i} - \pi_{t,i}^P), \quad |\tilde{Y}_{t,i}| \leq R_P.$$

The momentum variable is

$$g_t^P = \rho_{\text{mom}} \tilde{g}_t + (1 - \rho_{\text{mom}}) g_{t-1}^P,$$

and, under the interior tangent-projection condition stated in Assumption S1, the projected centered noise used in the P-MMF dual diagnostic is $Z_t^{P,\perp} = P_\perp \Gamma^{-1} \tilde{Y}_t$. This definition is an item-level stochastic analogue for comparison; it is not claimed to be an additional theorem about the original P-MMF implementation beyond the externally invoked P-MMF guarantee. The residual scale R_P is carried explicitly in Theorem 7; if a particular P-MMF implementation uses unit residuals, one may set $R_P = 1$.

Relation to the deployed deterministic policy. If the score vector has no exact ties, or if a deterministic tie-breaking rule is fixed, the deployed deterministic Algorithm 1 is the $\beta \rightarrow \infty$ limit of Algorithm 1-S because the perturbation $G_{t,i}/\beta$ vanishes. Theorems 6 and 7 analyze the finite-temperature stochastic relaxation Algorithm 1-S. They do not imply nonzero stochastic variance bounds for the deployed deterministic top- K policy; the deterministic policy is discussed only as the zero-temperature limit and is supported empirically by Figure 4 of the main paper.

We then calculate the variance of the martingale part of the projected P-MMF dual variable in Lemma 2.

Assumption S 1 (Interior tangent projection and non-degenerate projected P-MMF noise). Let $\Gamma := \text{diag}(\gamma_1, \dots, \gamma_I)$ and

$$P_{\perp} := I_I - \frac{1}{I} \mathbf{1} \mathbf{1}^{\top}$$

be the Euclidean projection onto the tangent subspace $\{z \in \mathbb{R}^I : \mathbf{1}^{\top} z = 0\}$. For the variance comparison with P-MMF, we make the following two assumptions.

1. The P-MMF projection is evaluated in a fixed relative interior of its simplex/resource constraint, so no non-negativity inequality is active. Equivalently, the stochastic part of the projected update is the linear tangent projection $P_{\perp}$. This is the precise condition under which a simplex projection reduces locally to subtracting a common average. Without this interior condition, the simplex projection is a thresholding map with an active set, not a global averaging map.
2. If $\tilde{Y}_t := \tilde{g}_t - \mathbb{E}[\tilde{g}_t \mid \mathcal{H}_t]$ denotes the primitive centered P-MMF sampling noise and

$$Z_t^{\text{P}, \perp} := P_{\perp} \Gamma^{-1} \tilde{Y}_t,$$

then for every item i there is a constant $\sigma_{\text{P},i}^2 > 0$ such that

$$\mathbb{E}[(Z_{t,i}^{\text{P}, \perp})^2 \mid \mathcal{H}_t] \geq \sigma_{\text{P},i}^2 \quad \text{for all } t.$$

We write $\sigma_{\text{P}}^2 := \min_i \sigma_{\text{P},i}^2 > 0$.

This assumption deliberately replaces the earlier informal claim that Gumbel-top- K coordinates are independent. Fixed-size top- K sampling creates cross-coordinate covariances; the theorem below only needs the projected conditional variance lower bound above. The lower bound is an assumption, not a consequence of Gumbel-top- K alone, and it can be verified from the conditional covariance matrix of the P-MMF sampler. A sufficient special case is independent Bernoulli exploration with conditional coordinate variances at least $\sigma_{\min}^{2,\text{P}}$, for which

$$\sigma_{\text{P}}^2 \geq \left(1 - \frac{1}{I}\right) \frac{\sigma_{\min}^{2,\text{P}}}{\gamma_{\max}^2}.$$

This sufficient case is only a comparison example and is not asserted for fixed-size Gumbel-top- K . For fixed-size Gumbel-top- K , σ_{P}^2 can instead be verified from the conditional covariance matrix $\Sigma_t := \mathbb{E}[\tilde{Y}_t \tilde{Y}_t^{\top} \mid \mathcal{H}_t]$ via $e_i^{\top} P_{\perp} \Gamma^{-1} \Sigma_t \Gamma^{-1} P_{\perp} e_i$.

Lemma 2 (P-MMF projected-noise variance lower bound). Under the P-MMF momentum update with $\rho_{\text{mom}} \in (0, 1)$ and Assumption S1, the martingale part of the projected P-MMF dual variable satisfies, for every item i ,

$$\text{Var}[\mu_{T+1,i}^{\text{P}, \perp}] \geq \eta^2 \sigma_{\text{P},i}^2 \kappa_T(\rho_{\text{mom}}) \geq \eta^2 \sigma_{\text{P}}^2 \kappa_T(\rho_{\text{mom}}),$$

where

$$\kappa_T(\rho_{\text{mom}}) := \sum_{r=1}^T \left(1 - (1 - \rho_{\text{mom}})^r\right)^2 = T - \frac{2(1 - \rho_{\text{mom}})}{\rho_{\text{mom}}} (1 - (1 - \rho_{\text{mom}})^T) + \frac{(1 - \rho_{\text{mom}})^2}{\rho_{\text{mom}}(2 - \rho_{\text{mom}})} (1 - (1 - \rho_{\text{mom}})^{2T}).$$

In particular,

$$\kappa_T(\rho_{\text{mom}}) \geq T - \frac{2(1 - \rho_{\text{mom}})}{\rho_{\text{mom}}}.$$

Proof. Let $q := 1 - \rho_{\text{mom}}$. The centered part of the P-MMF momentum recursion is

$$\bar{g}_t^{\text{P}} = \rho_{\text{mom}} \tilde{Y}_t + q \bar{g}_{t-1}^{\text{P}}, \quad \bar{g}_0^{\text{P}} = 0,$$

so induction gives

$$\bar{g}_t^{\text{P}} = \rho_{\text{mom}} \sum_{u=1}^t q^{t-u} \tilde{Y}_u.$$

By Assumption S1, the stochastic part of the projected dual update is obtained by applying the linear tangent projection $P_\perp$ to the resource-normalized noise. Hence

$$M_{T+1,i}^{\mathbf{P},\perp} := -\eta \sum_{t=1}^T (P_\perp \Gamma^{-1} \tilde{g}_t^{\mathbf{P}})_i = -\eta \sum_{u=1}^T \left(\rho_{\text{mom}} \sum_{t=u}^T q^{t-u} \right) (P_\perp \Gamma^{-1} \tilde{Y}_u)_i.$$

The inner geometric sum is

$$\rho_{\text{mom}} \sum_{t=u}^T q^{t-u} = 1 - q^{T-u+1}.$$

Therefore, with $Z_u^{\mathbf{P},\perp} := P_\perp \Gamma^{-1} \tilde{Y}_u$,

$$M_{T+1,i}^{\mathbf{P},\perp} = -\eta \sum_{u=1}^T (1 - q^{T-u+1}) Z_{u,i}^{\mathbf{P},\perp}.$$

Because $\tilde{Y}_u$ is a martingale-difference sequence and $P_\perp \Gamma^{-1}$ is deterministic, $Z_u^{\mathbf{P},\perp}$ is also a martingale-difference sequence. Thus cross-time covariances vanish by the tower property: for $u < v$,

$$\mathbb{E}[Z_{u,i}^{\mathbf{P},\perp} Z_{v,i}^{\mathbf{P},\perp}] = \mathbb{E}[Z_{u,i}^{\mathbf{P},\perp} \mathbb{E}[Z_{v,i}^{\mathbf{P},\perp} | \mathcal{H}_v]] = 0.$$

Consequently,

$$\begin{aligned} \text{Var}[M_{T+1,i}^{\mathbf{P},\perp}] &= \eta^2 \sum_{u=1}^T (1 - q^{T-u+1})^2 \mathbb{E}[(Z_{u,i}^{\mathbf{P},\perp})^2] \\ &= \eta^2 \sum_{u=1}^T (1 - q^{T-u+1})^2 \mathbb{E}[\mathbb{E}[(Z_{u,i}^{\mathbf{P},\perp})^2 | \mathcal{H}_u]] \\ &\geq \eta^2 \sigma_{\mathbf{P},i}^2 \sum_{u=1}^T (1 - q^{T-u+1})^2. \end{aligned}$$

Re-indexing $r = T - u + 1$ gives the stated lower bound with $\kappa_T(\rho_{\text{mom}}) = \sum_{r=1}^T (1 - q^r)^2$. Expanding the two geometric series gives the displayed closed form. Finally,

$$\kappa_T(\rho_{\text{mom}}) = T - 2 \sum_{r=1}^T q^r + \sum_{r=1}^T q^{2r} \geq T - 2 \sum_{r=1}^{\infty} q^r = T - \frac{2(1 - \rho_{\text{mom}})}{\rho_{\text{mom}}}.$$

This proves the lemma. Notice that the proof never assumes coordinate-wise independence of the fixed-size top- K indicators; all cross-coordinate effects are contained in the projected variance constant $\sigma_{\mathbf{P}}^2$. $\square$

Then we compare the martingale-noise components of the native dual-update variables in DivFair and P-MMF under the stated stochastic-relaxation assumptions.

Theorem 6 (Conditional native-unit martingale-noise diagnostic). *Under Algorithm 1-S and the P-MMF stochastic analogue, suppose Assumption S1 holds with $\sigma_{\mathbf{P}}^2 > 0$ and that both native updates use the same scalar learning rate η . Let*

$$c_{\rho,\text{mom}} := \frac{2(1 - \rho_{\text{mom}})}{\rho_{\text{mom}}}, \quad T_* := \frac{c_{\rho,\text{mom}} + \sqrt{c_{\rho,\text{mom}}^2 + 4\sigma_{\text{max}}^{2,\text{D}}/\sigma_{\mathbf{P}}^2}}{2}.$$

Then for every item i , whenever $T > T_*$,

$$\text{Var}[\mu_{T+1,i}^{\text{D}}] < \text{Var}[\mu_{T+1,i}^{\mathbf{P},\perp}], \quad (26)$$

where both variances refer to the martingale, stochastic-noise part of the corresponding dual variable.

Proof. We prove the result by comparing an upper bound on DivFair's martingale-noise variance with a lower bound on P-MMF's projected martingale-noise variance.

Step 1: Conditional variance of the DivFair gradient. Recall that $\mathcal{H}_t$ is the pre-sampling filtration containing the past decisions and the current user, preference vector, score vector, and mask. Algorithm 1 line 11 with $\boldsymbol{\mu}_t \geq \mathbf{0}$ gives $\mathbf{a}_t = \boldsymbol{\tau}$ deterministically, and line 13 uses $g_{t,i}^D = (\tau_i - x_{t,i})/T$. Since τ_i is deterministic and $\pi_{t,i}^D$ is $\mathcal{H}_t$ -measurable,

$$\mathbb{E}[g_{t,i}^D \mid \mathcal{H}_t] = \frac{\tau_i - \pi_{t,i}^D}{T}, \quad \text{Var}(g_{t,i}^D \mid \mathcal{H}_t) = \frac{\sigma_{t,i}^{2,D}}{T^2} \leq \frac{\sigma_{\max}^{2,D}}{T^2}.$$

Step 2: DivFair martingale-noise variance. Let

$$\xi_{t,i}^D := (\mu_{t+1,i}^D - \mu_{t,i}^D) - \mathbb{E}[\mu_{t+1,i}^D - \mu_{t,i}^D \mid \mathcal{H}_t].$$

Then $\{\xi_{t,i}^D\}$ is a martingale-difference sequence. Because the projection onto the box $\mathcal{U}_B = [0, B]^I$ is coordinate-wise clipping, each coordinate map is 1-Lipschitz. Conditional on $\mathcal{H}_t$, $\mu_{t,i}^D$ is fixed, and therefore

$$\text{Var}(\xi_{t,i}^D \mid \mathcal{H}_t) \leq \eta^2 \text{Var}(g_{t,i}^D \mid \mathcal{H}_t) \leq \frac{\eta^2 \sigma_{\max}^{2,D}}{T^2}.$$

Let $M_{T+1,i}^D := \sum_{t=1}^T \xi_{t,i}^D$. Orthogonality of martingale differences gives

$$\text{Var}[\mu_{T+1,i}^D] := \text{Var}(M_{T+1,i}^D) = \sum_{t=1}^T \mathbb{E}[\text{Var}(\xi_{t,i}^D \mid \mathcal{H}_t)] \leq \frac{\eta^2 \sigma_{\max}^{2,D}}{T}.$$

Step 3: P-MMF projected-noise lower bound. Lemma 2 and Assumption S1 give, for every item i ,

$$\text{Var}[\mu_{T+1,i}^{P,\perp}] \geq \eta^2 \sigma_P^2 \kappa_T(\rho_{\text{mom}}) \geq \eta^2 \sigma_P^2 (T - c_{\rho,\text{mom}}), \quad c_{\rho,\text{mom}} = \frac{2(1 - \rho_{\text{mom}})}{\rho_{\text{mom}}}.$$

Step 4: Variance comparison. Combining the two bounds,

$$\text{Var}[\mu_{T+1,i}^{P,\perp}] - \text{Var}[\mu_{T+1,i}^D] \geq \eta^2 \left[\sigma_P^2 (T - c_{\rho,\text{mom}}) - \frac{\sigma_{\max}^{2,D}}{T} \right].$$

The right-hand side is positive exactly when

$$\sigma_P^2 T^2 - \sigma_P^2 c_{\rho,\text{mom}} T - \sigma_{\max}^{2,D} > 0,$$

which holds for

$$T > \frac{c_{\rho,\text{mom}} + \sqrt{c_{\rho,\text{mom}}^2 + 4\sigma_{\max}^{2,D}/\sigma_P^2}}{2} = T_*.$$

This proves (26). $\square$

Remark. The comparison assumes the same scalar learning rate η for the two native updates. If different learning rates are used, the threshold condition should be replaced by $\eta_P^2 \sigma_P^2 (T - c_{\rho,\text{mom}}) > \eta_D^2 \sigma_{\max}^{2,D}/T$. The comparison is intentionally stated with the observable projected-noise constant σ_P^2 rather than with an independence-based closed form. This avoids treating the coordinates of a fixed-size top- K sampler as independent. In an implementation, σ_P^2 can be estimated or lower-bounded from the conditional covariance matrices of the stochastic P-MMF sampler. Under the sufficient independent-Bernoulli special case described in Assumption S1, one may use the conservative lower bound

$$\sigma_P^2 \geq \left(1 - \frac{1}{I}\right) \frac{\sigma_{\min}^{2,P}}{\gamma_{\max}^2}$$

inside T_* . For the deployed deterministic top- K algorithm, Theorem 6 should be read as an analysis of the finite-temperature stochastic relaxation Algorithm 1-S; empirical stability of the deterministic implementation is reported in Figure 4. The theorem compares only martingale-noise components in native update units. It does not bound the total movement, deterministic drift, or convergence rate of the full projected dual iterates, where predictable drift and active projection effects may dominate.

Concentration of centered native-update noise. In online optimization, the centered increment $Y_{t,i} = g_{t,i} - \mathbb{E}[g_{t,i} \mid \mathcal{H}_t]$ measures the stochastic noise in the dual update after removing the predictable exposure drift. Theorem 7 compares these centered fluctuations for finite-temperature DivFair Algorithm 1-S and its finite-temperature P-MMF analogue. It is a noise-concentration statement for averaged centered stochastic increments; it is not a claim that the uncentered exposure gap itself has zero bias, not a bound on the predictable drift component of the dual dynamics, and not a statement about nonzero stochastic variance under the deterministic top- K policy.

Lemma 3 (Freedman’s inequality). *Let $\{X_t\}_{t=1}^T$ be a martingale-difference sequence with respect to a filtration $\{\mathcal{G}_t\}_{t=0}^T$ and suppose $|X_t| \leq b$ almost surely. Define the predictable quadratic variation*

$$V_T := \sum_{t=1}^T \text{Var}(X_t \mid \mathcal{G}_{t-1}).$$

If $V_T \leq v$ almost surely for a deterministic constant $v > 0$, then, for any $\lambda > 0$,

$$\Pr \left[\sum_{t=1}^T X_t \geq \lambda \right] \leq \exp \left(-\frac{\lambda^2}{2(v + b\lambda/3)} \right).$$

Equivalently, the same bound holds on the event $\{V_T \leq v\}$ in the usual event form of Freedman’s inequality. Applying the bound to both X_t and $-X_t$ gives the corresponding two-sided inequality.

Theorem 7 (Averaged centered native-unit noise diagnostic). *Under finite-temperature Algorithm 1-S and its finite-temperature P-MMF analogue, define the centered sub-gradient increments*

$$Y_{t,i}^{\text{D}} := g_{t,i}^{\text{D}} - \mathbb{E}[g_{t,i}^{\text{D}} \mid \mathcal{H}_t], \quad Y_{t,i}^{\text{P}} := g_{t,i}^{\text{P}} - \mathbb{E}[g_{t,i}^{\text{P}} \mid \mathcal{H}_t].$$

Then, for every failure probability $\varepsilon \in (0, 1)$, with probability at least $1 - \varepsilon$ simultaneously for all items i ,

$$\left| \frac{1}{T} \sum_{t=1}^T Y_{t,i}^{\text{D}} \right| \leq C_{\text{D}} \left(\sqrt{\frac{\sigma_{\max}^{2,\text{D}} \log(4I/\varepsilon)}{T^3}} + \frac{\log(4I/\varepsilon)}{T^2} \right),$$

and

$$\left| \frac{1}{T} \sum_{t=1}^T Y_{t,i}^{\text{P}} \right| \leq C_{\text{P}}^{\text{conc}} \left(\rho_{\text{mom}} R_{\text{P}} \sqrt{\frac{\sigma_{\max}^{2,\text{P}} \log(4I/\varepsilon)}{T}} + \frac{\rho_{\text{mom}} R_{\text{P}} \log(4I/\varepsilon)}{T} \right),$$

for universal numerical constants $C_{\text{D}}, C_{\text{P}}^{\text{conc}} > 0$. Equivalently, before the outer averaging by $1/T$, DivFair’s centered cumulative native-update noise satisfies $|\sum_t Y_{t,i}^{\text{D}}| = O(T^{-1/2} \sqrt{\log(I/\varepsilon)})$ up to lower-order terms. In big- O notation, the averaged centered fluctuation is $O(T^{-3/2} \sqrt{\log(I/\varepsilon)})$ for DivFair’s native horizon-normalized update and $O(\rho_{\text{mom}} R_{\text{P}} T^{-1/2} \sqrt{\log(I/\varepsilon)})$ for the native P-MMF analogue, up to lower-order logarithmic terms. This is a native-update noise comparison; if P-MMF is also horizon-normalized by $1/T$, its displayed bound is rescaled accordingly.

Proof. We use the pre-sampling filtration $\mathcal{H}_t$ defined above and the post-sampling filtration $\mathcal{F}_t := \sigma(\mathcal{H}_t, \mathbf{x}_t)$. Equivalently, define the interleaved filtration $\mathcal{K}_{2t-1} = \mathcal{H}_t$ and $\mathcal{K}_{2t} = \mathcal{F}_t$. For each fixed item i , the variables $Y_{t,i}^{\text{D}}$ and $Y_{t,i}^{\text{P}}$ are $\mathcal{F}_t$ -measurable and satisfy $\mathbb{E}[Y_{t,i}^{\text{D}} \mid \mathcal{H}_t] = \mathbb{E}[Y_{t,i}^{\text{P}} \mid \mathcal{H}_t] = 0$ by construction. Therefore each fixed coordinate sequence is a martingale-difference sequence with respect to the interleaved filtration, and Freedman’s inequality applies to the sums over t .

For DivFair,

$$g_{t,i}^{\text{D}} = \frac{\tau_i - x_{t,i}}{T}, \quad \mathbb{E}[g_{t,i}^{\text{D}} \mid \mathcal{H}_t] = \frac{\tau_i - \pi_{t,i}^{\text{D}}}{T}.$$

Thus

$$|Y_{t,i}^{\text{D}}| = \frac{|x_{t,i} - \pi_{t,i}^{\text{D}}|}{T} \leq \frac{1}{T}, \quad \text{Var}(Y_{t,i}^{\text{D}} \mid \mathcal{H}_t) = \frac{\sigma_{t,i}^{2,\text{D}}}{T^2} \leq \frac{\sigma_{\max}^{2,\text{D}}}{T^2}.$$

Consequently $b_{\text{D}} = 1/T$ and

$$V_T^{\text{D}}(i) := \sum_{t=1}^T \text{Var}(Y_{t,i}^{\text{D}} \mid \mathcal{H}_t) \leq \frac{\sigma_{\max}^{2,\text{D}}}{T}.$$

For P-MMF, write its momentum recursion as

$$g_t^P = \rho_{\text{mom}} \tilde{g}_t + (1 - \rho_{\text{mom}}) g_{t-1}^P,$$

where g_{t-1}^P and the predictable quantities inside $\tilde{g}_t$ are $\mathcal{H}_t$ -measurable. By the explicit residual convention above,

$$\tilde{g}_{t,i} - \mathbb{E}[\tilde{g}_{t,i} \mid \mathcal{H}_t] = -r_{t,i}^P(x_{t,i} - \pi_{t,i}^P), \quad |r_{t,i}^P| \leq R_P.$$

Hence the centered instantaneous gradient is

$$Y_{t,i}^P = \rho_{\text{mom}}(\tilde{g}_{t,i} - \mathbb{E}[\tilde{g}_{t,i} \mid \mathcal{H}_t]) = -\rho_{\text{mom}} r_{t,i}^P(x_{t,i} - \pi_{t,i}^P).$$

Therefore

$$|Y_{t,i}^P| \leq \rho_{\text{mom}} R_P, \quad \text{Var}(Y_{t,i}^P \mid \mathcal{H}_t) \leq \rho_{\text{mom}}^2 R_P^2 \sigma_{t,i}^{2,P} \leq \rho_{\text{mom}}^2 R_P^2 \sigma_{\max}^{2,P},$$

and

$$V_T^P(i) := \sum_{t=1}^T \text{Var}(Y_{t,i}^P \mid \mathcal{H}_t) \leq \rho_{\text{mom}}^2 R_P^2 \sigma_{\max}^{2,P} T.$$

These bounds use only marginal Bernoulli variances for a fixed coordinate; they do not require coordinate-wise independence of the top- K indicators.

Apply Lemma 3 to each fixed coordinate and to both signs. A standard consequence is that, with probability at least $1 - \varepsilon/(2I)$ for a fixed item,

$$\left| \sum_{t=1}^T Y_{t,i} \right| \leq 2\sqrt{V_T(i) \log(4I/\varepsilon)} + \frac{2b \log(4I/\varepsilon)}{3}.$$

Using the DivFair values $V_T^D(i) \leq \sigma_{\max}^{2,D}/T$ and $b_D = 1/T$, then dividing by the outer averaging factor T , gives

$$\left| \frac{1}{T} \sum_{t=1}^T Y_{t,i}^D \right| \leq 2\sqrt{\frac{\sigma_{\max}^{2,D} \log(4I/\varepsilon)}{T^3}} + \frac{2 \log(4I/\varepsilon)}{3T^2},$$

which is the displayed DivFair bound up to the universal constant C_D . Using the P-MMF values $V_T^P(i) \leq \rho_{\text{mom}}^2 R_P^2 \sigma_{\max}^{2,P} T$ and $b_P = \rho_{\text{mom}} R_P$, then dividing by T , gives the displayed P-MMF bound. A union bound over both models and all I items, i.e., over $2I$ two-sided model-item events, yields the simultaneous probability $1 - \varepsilon$. $\square$

Theorem 7, together with Theorem 6, diagnoses the centered stochastic-noise component in the native dual-update variables for finite-temperature DivFair Algorithm 1-S and the corresponding native finite-temperature P-MMF analogue under the stated native-unit convention. This is a stochastic-relaxation diagnostic, not a same-unit deterministic convergence theorem and not a bound on total dual-iterate movement, predictable drift, or active projection effects. For P-MMF, Theorem 7 controls the centered per-round innovation sequence; concentration of the final momentum-accumulated dual noise would require applying Freedman's inequality to the geometrically weighted martingale sum appearing in Lemma 2. If P-MMF is rewritten with the same horizon normalization, the order separation is rescaled and only constant-level comparisons remain. For the deployed deterministic top- K algorithm, these results should be read as analysis of the stochastic relaxation; empirical stability of the deterministic implementation is reported in Figure 4.

Evaluation metrics and threshold definitions. The theory above uses a fixed objective threshold $\delta_{\text{obj}} \in (0, 1]$ during optimization. If an experiment reports coverage using a threshold computed from final utilities, that threshold is denoted separately by δ_{eval} and is used only for evaluation. The following formulas are the metric conventions used for reporting.

Unless otherwise stated, $\text{gain}_t(i) = w_{t,i}$ in score-retention computations. If held-out relevance labels are used instead, the gain function is stated explicitly. Coverage and Gini below are utility-based and use the final utility vector $\mathbf{v}_T$; exposure-based analogues should be reported separately as functions of $\bar{\mathbf{x}}$. If a denominator in a displayed ratio is zero, the metric is reported as undefined and excluded from averages unless an explicit small- ε denominator convention is declared; experiments should report the number or fraction of such undefined cases and whether macro- or micro-averaging is used.

Utility coverage. For a final utility vector $\mathbf{v}_T$,

$$\text{Coverage}_{\delta_{\text{eval}}}(\mathbf{v}_T) = \frac{1}{I} \sum_{i=1}^I \mathbf{1}\{v_{T,i} \geq \delta_{\text{eval}}\}.$$

Utility Gini. For nonnegative utilities with positive total mass,

$$\text{Gini}(\mathbf{v}_T) = \frac{\sum_{i=1}^I \sum_{j=1}^I |v_{T,i} - v_{T,j}|}{2I \sum_{i=1}^I v_{T,i}}.$$

If $\sum_i v_{T,i} = 0$, the metric is reported as undefined or computed with an explicitly stated small- ε denominator convention.

Relative DCG / utility retention. If $L_t^{\text{rerank}} = (i_1, \dots, i_K)$ is the reranked list and L_t^{base} is the original base-ranker list, define

$$\text{DCG@K}(L_t) = \sum_{r=1}^K \frac{\text{gain}_t(i_r)}{\log_2(r+1)}, \quad \text{Retention@K} = \frac{\sum_{t=1}^T \text{DCG@K}(L_t^{\text{rerank}})}{\sum_{t=1}^T \text{DCG@K}(L_t^{\text{base}})}.$$

This is a base-ranker-normalized retention score, not standard NDCG. If the aggregate base denominator is zero, the retention score is reported as undefined unless an explicit small- ε denominator convention is declared.

Standard NDCG. When held-out relevance labels are available,

$$\text{NDCG@K}(L_t) = \frac{\text{DCG@K}(L_t)}{\text{IDCG@K}_t},$$

where IDCG@K_t is the ideal DCG at round t computed from the held-out labels. If $\text{IDCG@K}_t = 0$, the per-round NDCG is reported as undefined or handled by the stated evaluation convention.

Cap utilization. With final average exposure $\bar{x}_i = T^{-1} \sum_{t=1}^T x_{t,i}$, for items with $\tau_i > 0$ define

$$\text{CapUtilization}_i = \frac{\bar{x}_i}{\tau_i}.$$

Items with $\tau_i = 0$ are reported separately as zero-cap items, and experiments should report summary statistics such as mean, maximum, and selected quantiles over the positive-cap items.

Mask activation and fallback. Let $A_t = \{i : \theta_{t,i} + 1/T \leq \tau_i\}$ be the unmasked item set. Report

$$\text{MaskActivationRate} = \frac{1}{T} \sum_{t=1}^T \mathbf{1}\{|A_t| < I\}, \quad \text{FallbackRate} = \frac{1}{T} \sum_{t=1}^T \mathbf{1}\{|A_t| < K\}.$$

If fallback is triggered, cap violation should be reported separately and Theorem 5 should not be cited as covering the fallback overflow loss.

Post-hoc diagnostics for the offline gap and drift. On small or medium instances where the offline integer program and the online-compatible comparator can be solved or accurately approximated, we recommend reporting

$$\hat{\Gamma}_T^o = \widehat{\mathcal{W}}_{\text{OPT}} - \widehat{\mathcal{W}}_{\text{comp}}^o, \quad \mathfrak{D}_{T,+}(\hat{\mathbf{y}}^c),$$

together with the prefix-discrepancy curve in (10). These quantities are diagnostics, not assumptions hidden inside Theorem 5; they indicate whether the realized instance is in a regime where the full-offline decomposition (15) is small.

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